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Multi-view contrastive and cluster-guided learning for author name disambiguation

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ABSTRACT

The presence of multiple authors with identical names in academic literature poses significant challenges for accurate paper retrieval. Moreover, to deliver more intelligent services, literature management systems must construct scholar profiles based on author data; however, name ambiguity undoubtedly complicates this process. The Author Name Disambiguation (AND) task aims to resolve this ambiguity by accurately attributing papers to the correct authors within these systems. In recent years, graph-based methods have emerged as the primary approach to addressing this issue, achieving remarkable results. Nevertheless, these methods face two main limitations. First, the initial graph constructed based on heuristic rules contains inherent noise, and existing models lack sufficient robustness to this noise. Second, prior research has largely overlooked the clustering effect exhibited by homogeneous papers, and most approaches decouple representation learning and clustering-based disambiguation into two separate stages. This decoupling prevents the dynamic refinement of cluster labels, resulting in suboptimal disambiguation outcomes. To overcome these obstacles, we propose an innovative Multi-view Contrastive and Cluster-Guided Learning (MCCG) framework for AND. First, we introduce a multi-view contrastive module to enhance the model's robustness against graph noise. Then, we present a cluster-guided module that harnesses the natural clustering of homogeneous papers to integrate representation learning with clustering-based disambiguation. This integration facilitates the learning of more discriminative representations and enables the dynamic refinement of highly reliable cluster labels, ultimately improving disambiguation performance. Extensive experimental results on four commonly used benchmark datasets demonstrate that MCCG is both effective and outperforms state-of-the-art models. Our code is available at https://github.com/lapixiaoxin/MCCG.

1. Introduction

Academic literature management systems such as Google Scholar, DBLP, and AMiner often index numerous authors with identical names, which leads to ambiguity in search results. For example, a query for "Wei Zhang" might return profiles corresponding to a computer science professor, an economist, or various other individuals from different disciplines. In addition, many platforms create scholar profiles based on author names to assist researchers in tracking the work of scholars they are interested in. However, name ambiguity remains a significant challenge. The Author Name Disambiguation (AND) task seeks to resolve these ambiguities, thereby enhancing the accuracy and intelligence of literature management systems.

In recent years, graph-based methods have demonstrated outstanding performance in the AND task (Cheng, Chen, Zhang, & Tang, 2024; Li et al., 2021; Luo et al., 2020; Xie, Liu, Wang, & Jia, 2022; Zhang, Zhang, Yao, & Tang, 2018), as illustrated in Fig. 1. These approaches construct graphs that capture relationships among papers to quantify their similarity and subsequently cluster them into homogeneous groups. However, this process leaves several important aspects unaddressed in existing graph-based AND methods: (1) a lack of robustness to noise (false edges) in the graph; (2) the decoupled process of representation learning and clustering-based disambiguation, which lacks the ability to dynamically refine cluster labels; and (3) an oversight of the clustering effect of homogeneous papers (i.e., papers written by the same author). Firstly, when constructing the initial graph using heuristic rules,

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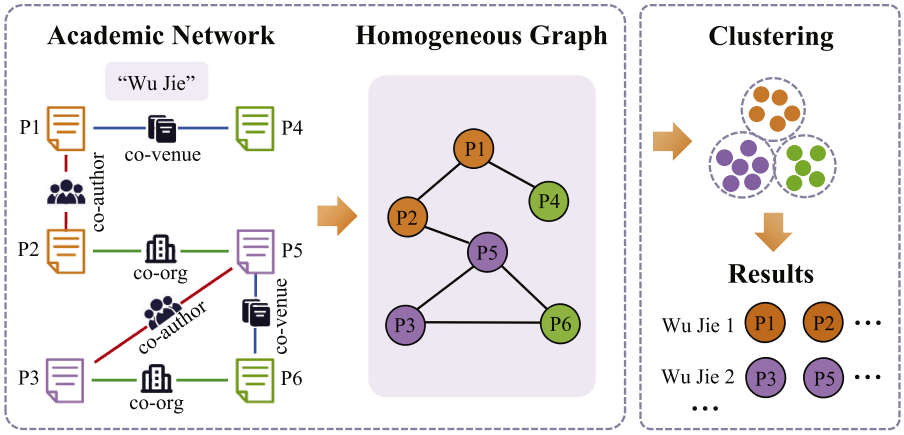


Fig. 1. Overview of the AND problem. P1 and P2 are homogeneous papers. Papers P2 and P5 are not actually authored by the same individual, therefore, the edge connecting them represents noise.

noise is inevitably introduced. Therefore, it is essential to improve model performance under noisy conditions. For example, in Fig. 1, an edge indicating a co-organization relationship exists between nodes P2 and P5, even though these nodes do not actually belong to the same author, this edge therefore represents noise. Existing methods have not effectively addressed such noise, leading to significant deficiencies in robustness. Secondly, the process of first performing representation learning to measure the similarity between pairs of paper embeddings and then applying a clustering algorithm for disambiguation is decoupled, lacking the ability to dynamically adjust cluster labels, which results in suboptimal clustering outcomes. Lastly, the clustering effect of homogeneous papers (represented by nodes of the same color in Fig. 1) aligns with the ultimate goal of the AND task. Papers by the same author should exhibit high similarity; however, this beneficial effect has been largely ignored in previous work. It is worth noting that this clustering effect is evident because the set of research areas for a given author does not change frequently, leading to a natural similarity among that author's papers.

The above analysis underscores the critical importance of enhancing the model's robustness to noise in the graph to achieve effective disambiguation. In this paper, we propose a multi-view cross contrastive learning (CL) approach. Specifically, we first apply adaptive graph augmentation techniques to randomly remove unimportant edges and embedding features. By introducing a controlled level of noise through graph perturbation, we generate two views that capture different contextual information. Then, guided by pseudo-label signals, we treat homogeneous paper samples within each view as positive pairs and the remaining samples as negative pairs, conducting cross-view CL. On one hand, the model learns to disregard the noise introduced by unimportant edges and focuses on the important patterns in the graph. On the other hand, cross-view CL maximizes the consistency of homogeneous paper embeddings at both the view and cluster levels. As a result, the model produces embedding representations that are more resilient to noise, ultimately improving disambiguation performance.

Furthermore, previous methods that treated representation learning and clustering-based disambiguation as a decoupled two-stage process lacked the ability to mutually reinforce embedding learning and disambiguation results, often leading to suboptimal outcomes. In this work, we leverage the clustering effect of homogeneous papers to seamlessly integrate representation learning and clustering tasks, and propose an adaptive sample-weighted clustering-guided module. This module enables the model to learn high-quality embeddings and dynamically optimize clustering results. Specifically, this module consists of the HDBSCAN clustering algorithm (McInnes & Healy, 2017) and an adaptive sample-weighted CL method, which also uses homogeneous papers as positive examples. The former is used to generate high-confidence pseudo-label guidance signals, while the latter learns discriminative paper representations and serves as input for the former, creating a synergistic effect. Traditional contrastive loss (Chen, Kornblith, Norouzi, & Hinton, 2020; van den, Li, & Vinyals, 2018) assigns equal weight to all sample pairs, limiting the model's ability to learn from hard samples (those that are difficult to distinguish) and correct false samples (those that are misclassified). We argue that sample pairs should contribute differently to the contrastive loss. To address this, we design a network that dynamically learns the weights of sample pairs, allowing the model to focus on different sample pairs to varying degrees during training. This approach enhances the model's ability to distinguish hard samples and correct misclassified ones. As a result, misclustered paper samples are given the opportunity to be corrected, mitigating the accumulation and propagation of erroneous guidance signals across multiple training epochs.

The combination of multi-view contrastive module and cluster-guided module enhances disambiguation performance, and the model is named MCCG. Extensive experiments on four widely used real-world datasets validate the effectiveness of MCCG.

In summary, our main contributions are as follows:

- We propose a multi-view contrastive module to enhance the model's robustness against noise in the graph, encouraging it to capture underlying semantic information from both views.
- We leverage the clustering effect of homogeneous papers to enhance the quality of embedding representations in both modules. Specifically, in the cluster-guided module, we utilize the clustering effect to unify representation learning and clustering task. The self-correcting ability of the cluster labels further enhances their collaborative effectiveness, jointly boosting AND performance.
- Experimental results on four widely used datasets demonstrate that our proposed model outperforms state-of-the-art methods. Extensive experiments further validate the effectiveness of each module within our model.

2. Related work

2.1. Author name disambiguation

The methods for solving AND task can be broadly categorized into three types based on the model: rule-based, machine learning-based, and GNN-based methods.

2.1.1. Rule-based methods

Rule-based methods primarily depend on manually defined criteria to evaluate the similarity between papers (Caron & van Eck, 2014; Lin

et al., 2017; Zhang & Ban, 2019; Zhang, Yu, Liu, & Wang, 2020). For example, if multiple co-authors are the same, it can be inferred that the ambiguous names likely refer to the same author entity. Similarly, if the ambiguous authors have the same organization and research focus across multiple papers, they may also be considered as the same entity. Cen, Dragut, Si, and Ouzzani (2013) defined rules based on five features, with potentially different embedding dimensions for each, performing pairwise similarity evaluation and proposing a hierarchical agglomerative clustering method with an adaptive stopping criterion for disambiguation. Louppe, Al-Natsheh, Susik, and Maguire (2016) manually defined 15 paper features (such as title, authors, affiliation, and keywords) and use rules such as Jaro-Winkler, cosine similarity, and character matching to assess the similarity between two papers. However, rule-based disambiguation methods exhibit two fundamental limitations: (1) they fail to capture critical relational patterns among papers, and (2) their dependence on manually crafted rules leads to inherent incompleteness, requiring ongoing refinement cycles.

2.1.2. Machine learning-based methods

Complementing rule-based approaches, machine learning-based methods constitute another major research paradigm in AND task (Han, Giles, Zha, Li, & Tsioutsiouliklis, 2004; Huang, Ertekin, & Giles, 2006; Louppe et al., 2016; Yoshida, Ikeda, Ono, Sato, & Nakagawa, 2010). These approaches typically integrate supervised learning frameworks with rule-based systems, operating through two principal phases: (1) textual feature extraction from paper components (including the abstract, title, and keywords) to create embedding representations, and (2) discriminative disambiguation using classification or clustering algorithms. For instance, Han et al. (2004) developed a Naive Bayes classifier to probabilistically model author-specific writing styles, while concurrently establishing a Support Vector Machine (SVM) framework for discriminative classification using labeled citation data. In a more recent approach, Rehs (2021) trained ensemble classifiers (Random Forest and Logistic Regression) with ground-truth author labels to differentiate between identity clusters. Nevertheless, these methods suffer from two principal constraints: (1) limited capacity for modeling deep semantic relationships, and (2) inability to adaptively capture evolving contextual patterns, resulting in suboptimal vector space representations that compromise disambiguation accuracy.

2.1.3. GNN-based methods

The advent of deep learning and graph neural networks (GNNs) has revolutionized embedding learning, with architectures including graph convolutional networks (GCN) (Kipf & Welling, 2016), graph attention networks (GAT) (Velickovic et al., 2017), and graph isomorphism networks (GIN) (Xu, Hu, Leskovec, & Jegelka, 2018) demonstrating exceptional representational power. These graph-based approaches for solving the AND task (Liu, Zhang, Chen, & Chen, 2024; Sun et al., 2020; Zhang et al., 2018; Zhou et al., 2024) typically construct attribute-rich paper graphs, then deploy GNNs to jointly encode node attributes and topological relationships, generating discriminative low-dimensional embeddings. This dual capability of capturing ego-network features and global structural patterns enables superior disambiguation accuracy compared to traditional heuristic methods, while supporting end-to-end optimization through differentiable learning. Pioneering privacy-preserving approaches, Zhang and Al Hasan (2017) developed anonymized graph structures through three carefully designed relational networks, preserving author confidentiality while maintaining disambiguation efficacy. Their framework optimizes node proximity using stochastic gradient descent, subsequently applying hierarchical agglomerative clustering (HAC) (Ward, 1963) for final grouping. MFAND (Zhou, Chen, Wang, Xu, & Zhao, 2021) decomposed the paper information and built six similarity graphs, implementing graph pruning during preprocessing to eliminate spurious edges before jointly encoding global and local patterns through its R3JG encoder, culminating in a binary classifier for authorship verification. BOND (Cheng et al., 2024) unified the decoupled task of AND into two stages: locally estimating paper similarities and globally clustering the papers. They proposed an end-to-end framework that enhances both the local similarity learning and the global clustering tasks through mutual reinforcement.

Beyond homogeneous graph approaches, recent AND research has evolved in two additional directions: (1) heterogeneous graph modeling and (2) cross-graph transformation techniques (Chen et al., 2023; Luo et al., 2020; Zheng, Zhang, Cui, Du, & Zhang, 2021). For example, MGATAND (Zhang et al., 2021) transformed academic heterogeneous graphs into multiple independent homogeneous graphs. GAT was employed to learn the embedding representations of each graph, optimizing a single embedding vector to jointly learn content and relational information. Pooja, Mondal, and Chandra (2022) proposed an attention-based multi-dimensional, multi-hop neighborhood graph convolutional network to capture multi-type relations and multi-neighborhood information in heterogeneous graphs, delving deeper into useful information from various types and neighbors. Wang et al. (2020) introduced a generative adversarial framework for AND. In this framework, the discriminative module differentiates whether two papers belong to the same author, while the generative module selects potentially homogeneous papers from the heterogeneous network, thus bypassing complex feature engineering. Both modules are trained collaboratively to enhance the model's ability to uncover latent relationships between papers. Li et al. (2021) proposed IUAD, a bottom-up incremental unsupervised AND method. It first constructs a stable collaboration network based on author collaboration relationships, then uses a probabilistic generative model to determine whether two papers in the network represent the same author. A global collaboration network is ultimately built, and for newly published papers, posterior probabilities are used to incrementally determine their author. HGNN-S (Huang, Zhang, Hao, Yang, & Wu, 2024) integrates semantic information from text, heterogeneous attributes, relationships, and neighborhood data through a pre-trained language model. Trained on a small amount of single-domain labeled data, HGNN-S performs cross-domain name disambiguation.

However, previous graph-based methods did not adequately consider that the construction of the initial graph inevitably introduces artificial noise, and these models do not explicitly enhance their robustness against such noise. Moreover, the inherent homogeneity of papers authored by the same individual has been overlooked in prior work. Additionally, decoupling representation learning from clustering-based disambiguation prevents dynamic optimization of the clustering results, ultimately yielding suboptimal disambiguation performance. The model proposed in this paper addresses these shortcomings.

2.2. Contrastive learning

In recent years, contrastive learning (CL) methods have shown exceptional performance in unsupervised learning (Chen et al., 2020) by generating discriminative embeddings through the objective of bringing positive samples closer and pushing negative samples apart. However, traditional CL methods (Chen et al., 2020; van den et al., 2018) often utilize only a single positive sample per anchor, which may introduce bias in clustering tasks where multiple similar samples exist within the same cluster. To address this, SupCon (Khosla et al., 2020) introduces a supervised CL method that treats samples from the same class as positive and samples from different classes as negative, thereby enhancing intra-class similarity and inter-class dissimilarity. Meanwhile, graph CL has rapidly developed (You et al., 2020; Zhu et al., 2021), providing strong support for capturing finer relational features. Inspired by these works, this paper employs graph CL methods to enhance the model's robustness against graph noise and to improve the quality of the learned embeddings.

3. Preliminary

3.1. Problem formulation

Given an author name a, which contains K authors with the same name, assume there are N papers belonging to name a, denoted as $P^a = \{p_1^a, p_2^a, \ldots, p_N^a\}$. Each paper p_i^a has multiple textual attributes, including title, abstract, keywords, author name and organization et al. These textual attributes are treated as Semantic Feature $S^a = \{s_1, s_2, \ldots, s_F\}$ or Relational Feature $R^a = \{r_1, r_2, \ldots, r_T\}$, where $s_f \in S^a$ denotes the f-th semantic feature (e.g., title/keywords); $r_t \in R^a$ denotes the t-th relational feature (e.g., co-author/co-organization); F and T represent the number of corresponding features.

3.1.1. Definition 1

Author Name Disambiguation. By leveraging semantic features S^a and relational features R^a, the AND task aims at finding a function Φ to partition P^a into a set of disjoint clusters C^a, i.e.,

$$\Phi(P^a|S^a, R^a) \rightarrow C^a, \text{where } C^a = \{C_1^a, C_2^a, \ldots, C_K^a\}, \tag{1}$$

where C_i^a denotes the homogeneous paper subset authored by the i-th author named a, here, $C_i^a \subset P^a$, $C_i^a \cap C_j^a = \emptyset$ and $\bigcup_{i=1}^{K} C_i^a = P^a$. For papers p_i^a and p_j^a authored by same individual in the same cluster, we have $\mathbb{I}(p_i^a) = \mathbb{I}(p_j^a), \forall (p_i^a, p_j^a) \in C_k^a \times C_k^a$, and for papers authored by different individuals in different clusters, $\mathbb{I}(p_i^a) \neq \mathbb{I}(p_j^a), \forall (p_i^a, p_j^a) \in C_k^a \times C_{k'}^a, k \neq k'$. $\mathbb{I}(p_i^a)$ is the author identification of the paper p_i^a. *For convenience, all the superscripts a will be omitted in later use.*

3.1.2. Definition 2

Homogeneous paper. For papers p_i and p_j, if $\mathbb{I}(p_i) = \mathbb{I}(p_j)$, they are referred to as homogeneous papers. That is, in the pseudo-label list $L = \{l_1, l_2, \ldots, l_N\}$ corresponds to the labels of the papers $P = \{p_1, p_2, \ldots, p_N\}$, if $l_i = l_j$, the papers are considered homogeneous.

3.2. Homogeneous graph

We build homogeneous graphs to address the AND task. A homogeneous graph is defined as follows.

3.2.1. Definition 3

Homogeneous graph. The homogeneous graph under name reference a is defined as $G = \{V, E\}$, where $V = \{v_1, v_2, \ldots, v_N\}$ is the paper vertex set and $E = \{e_{ij}\} \subseteq V \times V$ is the edge set. Use Γ_v to represent the set of vertex types and Γ_e to represent the set of edge types. In the homogeneous graph, $|\Gamma_v| + |\Gamma_e| = 2$.

4. Methodology

As illustrated in Fig. 2, we propose the MCCG framework for AND. This framework integrates several modules to enhance performance. Initially, the graph construction and augmentation module is responsible for creating the initial graph and generating views with different contextual information (Section 4.1). Next, the graph encoder network performs feature encoding on these multiple views (Section 4.2). Then, the multi-view contrastive module enhances the model's robustness against graph noise through a multi-view cross CL method (Section 4.3). Simultaneously, the cluster-guided module uses an adaptive sample-weighted CL to learn discriminative embeddings and generate high-confidence pseudo-label guidance signals (Section 4.4). Each module will be discussed in detail next.

4.1. Graph construction and augmentation

4.1.1. Initial graph construction

Literature management systems contain a vast number of papers. Constructing an initial graph based on all papers not only incurs significant computational overhead but also degrades the performance of disambiguation. In this paper, we construct separate graphs for each ambiguous name. We treat papers as vertices V and use three types of integrated relationships as edges E, including co-author (same name authors except for disambiguation authors), co-organization (affiliated with the same institution), and co-venue (appearing the same conference or journal). The edge construction rules employ different measurement scales and the detailed definition of each is as follows:

Co-author. Each paper includes an author list that contains multiple co-authors. For an ambiguous name a, assume that in two papers p_i and p_j where this name appears, the sets of co-author names (excluding a) are denoted by A_i and A_j, respectively. We treat the names as strings

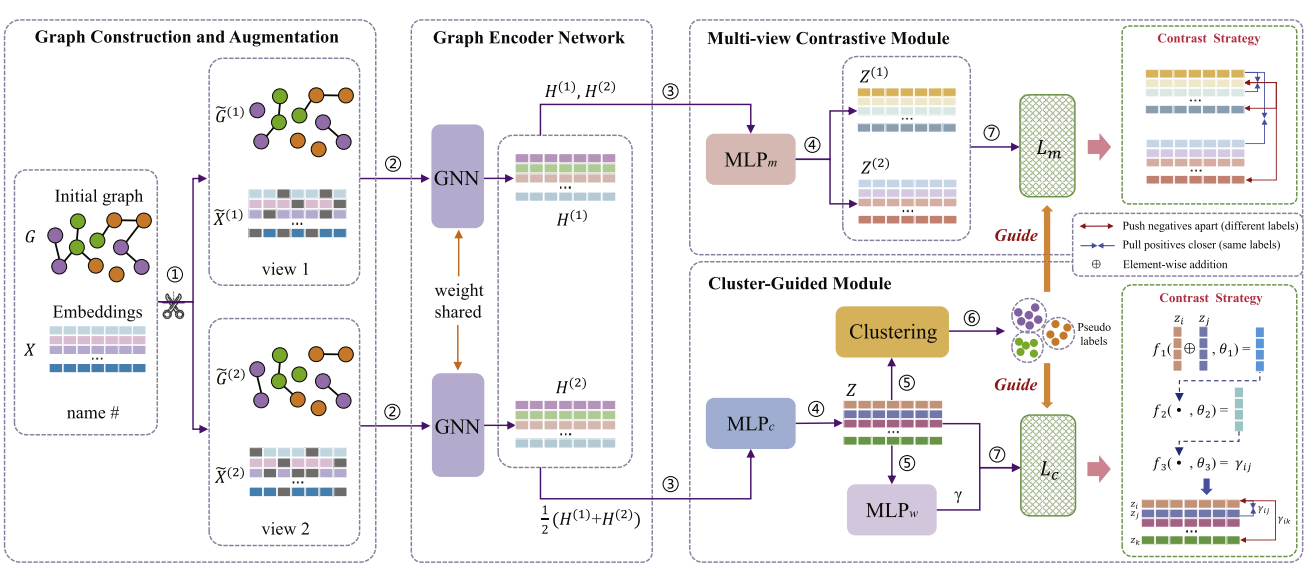


Fig. 2. The overall framework of MCCG. Firstly, graph augmentation techniques are applied to the initial graph to obtain two views. Then, these views are input into a graph encoder with shared weights to obtain two embeddings. Subsequently, both embeddings are passed into the Multi-view Contrastive (MC) module for cross-view CL. Simultaneously, the average of the two embeddings is provided to the Cluster-Guided (CG) module. This serves as input for the clustering algorithm to generate pseudo-labels that guide the MC and CG modules for CL. Additionally, it is used as input for our designed MLP to dynamically compute the weights of CG module sample pairs during CL.

and apply normalization (e.g. converting "Wang Minqiang" to "minqinangwang"). The similarity between the two papers is then calculated based on the count of *word overlap*, which is defined as $|A_i \cap A_j|$.

Co-venue. Papers published in the same conference or journal usually focus on similar research topics. We first convert the relevant text to lowercase, remove stopwords, and then tokenize the text based on whitespace to obtain the word sets W_i and W_j for papers p_i and p_j. The similarity between the two papers is then measured by the *word overlap*, defined as the cardinality of the intersection, i.e., $|W_i \cap W_j|$.

Co-organization. We extract the organization information associated with ambiguous names from the papers, convert the extracted strings to lowercase, remove stopwords, and then tokenize them based on whitespace to obtain the corresponding word sets O_i and O_j for papers p_i and p_j. Given that the organization information is generally noisier, we employ the *Jaccard Index* to evaluate similarity, which is defined as $\frac{|O_i \cap O_j|}{|O_i|+|O_j|-|O_i \cap O_j|}$.

An edge is established in the graph if any relationship between a pair of papers exceeds a predefined threshold. *Thus, in the final homogeneous graph, only one type of edge is retained to represent the connectivity between nodes.* As a result, the initial graph $G = \{V, E\}$ is constructed.

In addition to capturing the graph structure, the initial embeddings of the nodes play a crucial role. This paper uses title, keywords, and organization information as the semantic features. These features are then encoded to yield embedding representations X. We extract data including the title, abstract, keywords, organization, venue, and year from the entire dataset. After converting the extracted text to lowercase and removing erroneous characters and stopwords, the resulting data is used as the training corpus for the semantic encoder. We employ Word2Vec (Mikolov, Chen, Corrado, & Dean, 2013) as the semantic feature encoder owing to its efficient processing speed and robust performance. Further discussion on the semantic feature encoder is provided in Section B.5.

4.1.2. Adaptive graph augmentation

The heuristic rules employed to construct the initial graph frequently introduce noise that impairs the model's capacity to learn critical patterns. Since noise is unavoidably introduced during the graph construction process, the emphasis should be on enhancing the model's robustness against such noise to mitigate its adverse impact on performance. Our observations reveal that the noise present in the graph often exhibits distinct characteristics. For example, an edge might exist between two nodes even though their attribute information differs significantly, thereby indicating a weak correlation. Such spurious edges are likely to represent noise, and their presence can impede the model's ability to capture essential information from the graph.

In recent years, graph CL has attracted considerable attention owing to its remarkable performance in unsupervised learning through the use of graph augmentation techniques (You et al., 2020; Zhu et al., 2021). Motivated by these advances, we adopt the adaptive graph augmentation technique proposed in GCA (Zhu et al., 2021), which simultaneously improves both the graph topology and node attributes. Specifically, it first computes the importance of both edges and node attributes within the graph. Then, based on the predefined edge drop rate $r_{e,v}$ and feature drop rate $r_{f,v}$ (with $v = \{1, 2\}$), it introduces a controlled amount of noise by removing less important edges and node attributes, all while preserving the essential structure of the original graph. As a result, two augmented views are generated, denoted as $\{\widetilde{G}^{(v)}, \widetilde{X}^{(v)}\}$ for $v = \{1, 2\}$, where $\widetilde{G}^{(v)}$ is the augmented graph derived from the original graph G and $\widetilde{X}^{(v)}$ is the corresponding augmented embedding derived from X. It is worth noting that the information in these two views is not identical; in one view (i.e., $\{\widetilde{G}^{(1)}, \widetilde{X}^{(1)}\}$) a portion of the noise may be eliminated, whereas in the other view (i.e., $\{\widetilde{G}^{(2)}, \widetilde{X}^{(2)}\}$) some noise may still persist. Subsequently, by employing the method proposed later, the model can leverage these differences to capture the underlying semantics and patterns, thereby reducing the interference caused by noise.

4.2. Graph encoder network

To effectively fuse semantic and relational features, existing methods typically employ graph neural networks (GNN) (Kipf & Welling, 2016; Velickovic et al., 2017) to encode these features as low-dimensional embeddings. Similarly, we employ a variant of the GAT (referred to as GAT-v) (Wang et al., 2019). Unlike GAT, which only attends to one-hop neighbors, GAT-v utilizes higher-order neighbors to capture richer relational information. Node embeddings are aggregated based on attention scores from neighboring nodes. The edge weight calculation is as follows:

$$w_{ij} = a^T([Wx_i \,||\, Wx_j]), j \in N_i, \tag{2}$$

where N_i denotes the neighbor set of node i, x_i denotes an embedding in $\widetilde{X}$, W is a shared weight matrix, a is a weight vector for projecting the high-dimension to a real number and || is the concatenation operator. The attention coefficients is calculated as follows:

$$\alpha_{ij} = softmax_j(w_{ij}) = \frac{\exp(w_{ij})}{\sum_{k \in N_i} \exp(w_{ik})}, \tag{3}$$

GAT-v obtains a proximity matrix by considering t-order neighbor nodes in the graph:

$$M = (B + B^2 + \cdots + B^t)/t, \tag{4}$$

here B is the transition matrix where $B_{ij} = 1/d_i$ if $e_{ij} \in E$ and $B_{ij} = 0$ otherwise. d_i is the degree of node i. M_{ij} denotes the topological relevance of node j to node i up to t hops. t could be flexibly adjusted according to the actual situation. The attention coefficients after adding the topological weights M are as follows:

$$\alpha_{ij} = \frac{\exp(\delta(M_{ij} w_{ij}))}{\sum_{k \in N_i} \exp(\delta(M_{ik} w_{ik}))}, \tag{5}$$

where δ is an activation function (here LeakyReLU is used). We stack two graph attention layers to get the paper embeddings:

$$H'' = \sigma(A'' W''(\sigma(A' W' \widetilde{X}))), \tag{6}$$

where $A' = \{\alpha_{ij}\}$ and A'' represent the attention matrices, W' and W'' denote the projection matrices, σ is an activation function (here ELU is used).

In this way, GAT-v encoder encodes the embedding $\widetilde{X}$ into hidden embeddings $H = H''$. Similarly, the embedding of the two views is generated, denoted as $H^{(1)}$ and $H^{(2)}$.

4.3. Multi-view contrastive module

As discussed above, the initial graph inevitably contains noise that can degrade representation learning and, as a result, impair disambiguation performance. To enhance the model's robustness against noise, we propose a multi-view cross CL method, as illustrated in Fig. 3. In traditional graph CL (You et al., 2020; Zhu et al., 2021), given a graph with N nodes, graph augmentation generates two augmented views per node, resulting in a total of $2N$ augmented nodes. The representations corresponding to the two augmented views of the same node form a positive pair, while the representations from the other $2N - 2$ augmented nodes are regarded as negative pairs. Although this approach can achieve modest performance gains, it does not align with the objectives of the AND task. Specifically, our goal is to ensure that the embeddings of papers within the same cluster C_k are highly similar, whereas those of papers from different clusters exhibit lower similarity. However, the traditional contrastive loss fails to incorporate this clustering criterion, potentially limiting its effectiveness in enhancing model robustness. Noticing that homogeneous papers in the AND task exhibit clustering effect and inspired by SupCon (Khosla et al., 2020), we propose multi-view cross CL, treating homogeneous papers within each view as positive samples and heterogeneous papers as negative samples. This approach enables the model to extract richer latent representations from multiple views while

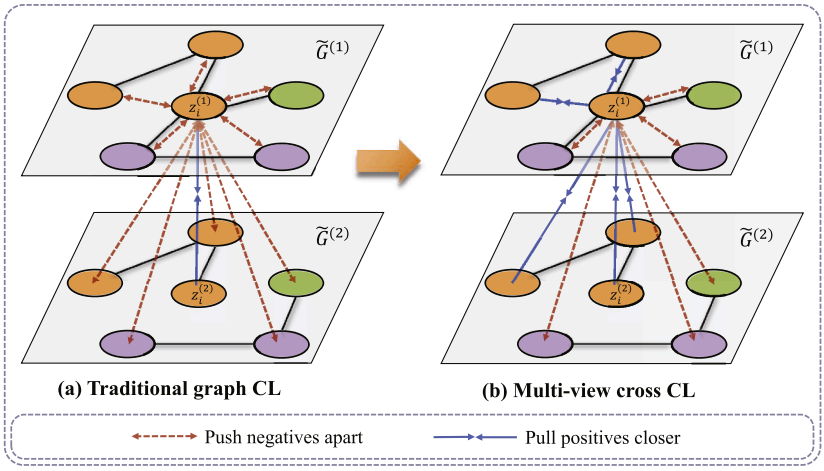


Fig. 3. Traditional graph CL vs. Multi-view cross CL. (a) In traditional graph CL, each sample $z_i^{(1)}$ has only one positive sample $z_i^{(2)}$, and the remaining samples in both views are considered negative. Contrastive learning is performed by pulling the positive sample pairs closer and pushing the negative pairs apart. (b) Multi-view cross CL leverages the clustering effect of homogeneous papers. For each sample $z_i^{(1)}$, all its homogeneous samples are treated as positive samples, while the remaining samples are considered negative. This approach extracts richer semantic information at both the view and cluster levels, thereby enhancing the robustness of the model to noise.

effectively capturing both intra-cluster consistency and inter-cluster differences. By enforcing consistency of embeddings across both views and clusters, the adverse effects of noise are effectively mitigated.

Specifically, we first feed the embeddings $H^{(1)}$ and $H^{(2)}$ from the two views generated by the graph encoder into the projection head MLP_m as shown in Fig. 2, which consists of two fully connected layers with ELU activation, to obtain the enhanced paper embeddings $Z^{(1)}$ and $Z^{(2)}$, i.e.,

$$Z^{(v)} = MLP_m(H^{(v)}, \phi_m), v \in \{1, 2\}, \tag{7}$$

where $Z^{(v)} = \{z_1^{(v)}, z_2^{(v)}, \ldots, z_n^{(v)}\}$ and ϕ_m represents the parameters of the MLP_m.

Then, by leveraging the pseudo-labels L provided by the cluster-guided module, we select positive and negative sample pairs for CL. Thus, for each paper sample i, the cross-view contrastive loss is defined as follows:

$$\mathcal{L}_i = -\frac{1}{|C_i|} \sum_{j \in C_i} \sum_{\alpha,\beta=1}^{2} \log \frac{\exp(s(z_i^{(\alpha)}, z_j^{(\beta)})/\tau_1)}{\sum_{\alpha',\beta'=1}^{2} \sum_{k \in Q_i} \exp(s(z_i^{(\alpha')}, z_k^{(\beta')})/\tau_1)}, \tag{8}$$

where C_i denotes the set of paper embeddings that belong to the same cluster as the i-th paper embeddings, and $|C_i|$ is its cardinality; $z_i^{(\alpha)}$ denotes the i-th row of embeddings $Z^{(v)}, v \in \{1, 2\}$; Q_i is the set of all paper embeddings except the i-th embedding; τ_1 is the temperature parameter. The similarity between the two papers is calculated using the cosine similarity, denoted as $s(\cdot, \cdot)$, which it is defined as:

$$s(z_i, z_j) = \frac{z_i^T \cdot z_j}{||z_i|| \cdot ||z_j||}, \tag{9}$$

Subsequently, considering N paper samples associated with the ambiguous name a, the final cross-view contrastive loss is defined as follows:

$$\mathcal{L}_m = \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}_i. \tag{10}$$

4.4. Cluster-guided module

Prior research has typically decomposed the AND task into a two-stage framework consisting of representation learning followed by clustering-based disambiguation. Compared with end-to-end paradigms, this decoupled approach often demonstrates compromised performance due to error propagation between stages. Recent advancements (Cheng et al., 2024; Zhou et al., 2024) have explored end-to-end solutions that leverage the self-supervised learning capabilities of neural networks to implicitly improve clustering outcomes. Nevertheless, these methods struggle with ambiguous decision boundaries and error correction, making them prone to local optima and reinforcement of incorrect clustering decisions. To address these limitations, this work proposes a cluster-guided module that seamlessly integrates representation learning with clustering-based disambiguation by exploiting the natural aggregation of homogeneous papers. This module not only facilitates the learning of discriminative embeddings but also generates high-confidence pseudo-labels to guide the disambiguation process. Moreover, its self-correcting mechanism enhances clustering robustness, leading to improved disambiguation performance.

This module consists of an adaptive sample-weighted CL method and the HDBSCAN (McInnes & Healy, 2017) clustering algorithm, both of which create a synergistic effect between representation learning and the clustering task. Specifically, we first average the embeddings $H^{(1)}$ and $H^{(2)}$, and then feed the result into a projection head MLP_c, as shown in Fig. 2. Note that MLP_c has the same structure as MLP_m, but uses independent weights, to obtain Z, i.e.,

$$Z = MLP_c(\frac{1}{2}(H^{(1)} + H^{(2)}), \phi_c), \tag{11}$$

where $Z = \{z_1, z_2, \ldots, z_n\}$ and ϕ_c represents the parameters of the MLP_c.

Secondly, we feed Z into HDBSCAN to obtain pseudo-labels L for the paper clusters, which serve as the guidance signals for both the cluster-guided module and the multi-view contrastive module.

Then, the core idea of the adaptive sample-weighted CL method is that different samples should be assigned different levels of attention during CL. Specifically, positive sample pairs with high similarity are expected to be assigned lower weights, whereas negative sample pairs with high similarity should receive higher weights, thereby encouraging the model to focus on hard samples and potentially false samples. Therefore, we define a three-layer perceptron MLP_w as shown in Fig. 2 to dynamically learn the weights of paper sample pairs, enabling the model to distinguish hard samples and correct false labels during training, thereby mitigating error accumulation. The weight γ_{ij} between papers p_i and p_j is computed as follows:

$$\gamma_{ij} = f_3(f_2(f_1(\mathcal{H}(z_i, z_j); \theta_1); \theta_2); \theta_3), \tag{12}$$

where $\mathcal{H}(z_i, z_j)$ is a pooling function and we use the SUM function, f_1, f_2, f_3 denote the network functions of each layer respec-

tively, $\theta_1, \theta_2, \theta_3$ are the corresponding parameters. The network function MLP_w is defined as follows:

$$f_1(h_1;\theta_1) = \sigma_1(Norm_1(W_1 h_1 + b_1)), \tag{13}$$

$$f_2(h_2;\theta_2) = \sigma_2(Norm_2(W_2 h_2 + b_2)), \tag{14}$$

$$f_3(h_3;\theta_3) = \sigma_3(W_3 h_3 + b_3), \tag{15}$$

which θ_i is all the parameters of each layer of MLP include learnable parameter matrix W_i, the bias vector b_i, and batch normalization parameters of $Norm_i$, $i = \{1,2,3\}$, $Norm$ is the normalization operation. σ_i is an activation function, σ_1 and σ_2 are the Tanh function, σ_3 is the sigmoid function. Based on the pseudo-labels L, samples in the embedding Z that share the same label are treated as positive samples, while the remaining samples are regarded as negative samples. Thus, the cluster-guided contrastive loss is defined as follows:

$$\mathcal{L}_c = -\frac{1}{N}\sum_{i=1}^{N}\frac{1}{|C_i|}\sum_{j\in C_i}\log\frac{\gamma_{ij}\exp(s(z_i,z_j)/\tau_2)}{\sum_{k\in Q_i}\gamma_{ik}\exp(s(z_i,z_k)/\tau_2)}, \tag{16}$$

where τ_2 is the temperature parameter; γ_{ij} is the weight coefficient between paper samples p_i and p_j; C_i denotes the set of paper embeddings that belong to the same cluster as the i-th paper embeddings, and $|C_i|$ is its cardinality; Q_i is the set of all paper embeddings except the i-th embedding; N represents the number of papers associated with the ambiguous name a.

4.5. Joint objective optimization

Finally, we combine the loss functions of the two modules and introduce a parameter ω to balance the two types of losses. The overall optimization objective of the model is expressed as:

$$\mathcal{L} = \omega\mathcal{L}_c + (1-\omega)\mathcal{L}_m. \tag{17}$$

4.6. Clustering

After learning the paper embeddings through the model proposed earlier, for each ambiguous name a, we apply a clustering algorithm to partition N papers into K disjoint clusters, where each cluster corresponds to an actual author. In this paper, we conduct transductive learning on the graph, where pseudo-labels generated in the final training epoch are treated as the ultimate disambiguation outcomes. We employ HDBSCAN (McInnes & Healy, 2017) as our clustering algorithm, which is a density-based method that does not require predefining the number of clusters. Compared with the hierarchical agglomerative clustering (HAC) algorithm (Ward, 1963), which necessitates predefining the number of clusters, HDBSCAN is more suitable for practical literature management system scenarios. Furthermore, in our proposed model, the clustering algorithm supplies guidance signals for contrastive learning via pseudo-labels generated during training, indicating that its performance has a significant impact on the overall training process. To further investigate the performance of alternative clustering algorithms on the AND task, we have experimented with several density-based clustering methods, the results of which are detailed in Appendix Table B.15. Thus, the complete framework of MCCG is presented in Algorithm 1.

Algorithm 1 The proposed framework: MCCG.

Input: Initial graph G^a and paper embedding X^a of name a.
Output: C^a, the disambiguation result of name a.
Initialize MCCG model with parameters θ.
Get the views $\{\widetilde{G}^{(1)}, \widetilde{X}^{(1)}\}$ and $\{\widetilde{G}^{(2)}, \widetilde{X}^{(2)}\}$ with GCA for graph augmentation.
for $iter = 1, 2, \ldots, T$ **do**
  Get hidden embeddings $H^{(1)}$ and $H^{(2)}$ with Eq. (6) via GAT-v encoder.
  Get embeddings $Z^{(1)}$ and $Z^{(2)}$ with Eq. (7) on $H^{(1)}$ and $H^{(2)}$.
  Get embedding Z with Eq. (11) on $\frac{1}{2}(H^{(1)} + H^{(2)})$.
  Get cluster-guided signal L with HDBSCAN on Z.
  Calculate the weights γ of pairwise samples in the embedding Z with Eq. (12) to subsequently compute cluster-guided contrastive loss $\mathcal{L}_c$.
  Calculate cross-view contrastive loss $\mathcal{L}_m$ with Eq. (10) and cluster-guided contrastive loss $\mathcal{L}_c$ with Eq. (16) based on the cluster-guided signal L.
  Calculate the joint loss $\mathcal{L}$ as the weighted sum of $\mathcal{L}_m$ and $\mathcal{L}_c$.
  Update the model parameters θ according to Eq. (17) and Adam optimizer.
end for
Pass the initial graph G^a and paper embedding X^a through the model with parameters θ to obtain the target embedding Z.
Perform HDBSCAN algorithm based on representation result Z.
return C^a.

5. Experiments

5.1. Experimental setup

5.1.1. Datasets

Four datasets are used to evaluate our model: three commonly used manually labeled large datasets, AMiner-AND[1], WhoisWho-v1[2], and WhoisWho-v2[2], and a large-scale dataset, LAGOS-AND[3], constructed based on authoritative academic resources, ORCID and DOI. AMiner-AND contains 600 author names, with 400 in the training set, 100 in the validation, and 100 in the test set. WhoisWho-v1 contains 321 names, with 221 in the training set, 50 in the validation set, and 50 in the test set. WhoisWho-v2 contains 165 names, with 101 in the training set, 30 in the validation set, and 34 in the test set. LAGOS-AND (Zhang, Lu, & Yang, 2023) contains 24,510 names, with 19,755 in the training set, 2314 in the valid set, and 2441 in the test set. Detailed information is provided in Table 1.

5.1.2. Baselines

To evaluate the performance of MCCG, we compare it with several existing AND models. For a fair comparison, all models are evaluated as single models without employing common post-processing techniques to enhance performance, including SND-all (Chen et al., 2023) and BOND (Cheng et al., 2024), which utilize *Tanimoto distance* for word matching to handle outliers. The baseline models are as follows:

- **AGAND** (Zhang & Al Hasan, 2017) takes privacy concerns in real-world scenarios into account by using only relational data to construct an anonymous graph for feature encoding, and employs HAC for clustering and disambiguation.
- **AMiner** (Zhang et al., 2018) integrates global metric learning with local linkage learning to jointly enhance disambiguation performance. The global metric learning component optimizes the spatial arrangement of embeddings by constructing positive and negative pairs and applying triplet loss, while local linkage learning further refines the embeddings via a graph reconstruction loss. Finally, hierarchical agglomerative clustering (HAC) is employed to perform the disambiguation clustering.
- **PHNet** (Qiao, Du, Fu, Wang, & Zhou, 2019) proposes a heterogeneous graph convolutional network (HGCN) to encode graph structural features and node attribute information, and introduces a GHAC method for disambiguation that does not require specifying the final number of clusters.
- **LAND** (Santini et al., 2022) generates embeddings using multimodal textual information from knowledge graphs, initializing entity embeddings with BERT (Devlin, 2018), and employing the knowl-

[1] https://www.aminer.cn/na-data.
[2] https://www.aminer.cn/billboard/whoiswho.
[3] https://zenodo.org/records/7313380.

Table 1
The detailed statistics of the four datasets.

	AMiner-AND	WhoisWho-v1	WhoisWho-v2	LAGOS-AND
# of names	600	321	165	24,510
# of authors	39,781	26,093	10,479	68,366
# of papers	208,827	292,448	167,144	758,584
# of authors per name min/max/avg	2/542/66.3	0/588/81.3	3/483/63.5	2/89/2.8
# of papers per name min/max/avg	192/916/348.0	0/5682/911.1	99/6142/1013.0	2/1067/30.9

edge representation learning method of LiteralE (Kristiadi, Khan, Lukovnikov, Lehmann, & Fischer, 2019). Finally, HAC is used for clustering-based disambiguation.

- **DHGN** (Zheng et al., 2021) uses fastText (Joulin, Grave, Bojanowski, & Mikolov, 2016)to encode node features in a heterogeneous graph and employs Hin2Vec (Fu, Lee, & Lei, 2017) to obtain relational embeddings for papers in the heterogeneous graph. Finally, DBSCAN is used for clustering.
- **MRAND** (Pooja et al., 2022) proposes an attention-based multi-dimensional, multi-hop neighborhood-based graph convolutional network for representation learning. It extracts relational information from both the relationship level and neighborhood level within the network, and ultimately applies HAC for clustering.
- **SND-all** (Chen et al., 2023) extracts semantic and relational features from papers, using Word2Vec to encode semantic features and metapath2vec to encode relational features. The density-based clustering algorithm DBSCAN (Ester, Kriegel, Sander, Xu et al., 1996) is used for disambiguation.
- **BOND** (Cheng et al., 2024) leverages both local and global information in an end-to-end manner to mutually enhance the training of the model. Local paper similarities drive the global clustering process, generating pseudo-labels, which in turn refine the local pairwise representations. Then, it also uses DBSCAN for clustering.
- **EAND** (Zhou et al., 2024) constructs six similarity graphs based on different similarity coefficients, raw document features, and the fusion feature. It also proposes a feature extraction model composed of three mixed attention mechanism layers to extract key information from the papers. Finally, the AND task is treated as a binary classification task, where the output probability is used to determine whether a pair of papers belongs to the same author.

5.1.3. Evaluation metrics

As in previous studies (Chen et al., 2023; Cheng et al., 2024; Zhang et al., 2018), three metrics are employed for the evaluation of the disambiguation effect of each name: pairwise precision Pre, pairwise recall Rec, and pairwise F1-score $F1$. The formulations of these evaluation metrics are as follows:

$$Pre = \frac{TP}{TP + FP}, \tag{18}$$

$$Rec = \frac{TP}{TP + FN}, \tag{19}$$

$$F1 = \frac{2 \cdot Pre \cdot Rec}{Pre + Rec}, \tag{20}$$

TP (True Positives) represents instances where the model correctly identifies two papers as being authored by the same person. FP (False Positives) represents instances where the model incorrectly identifies two papers as being authored by the same person. FN (False Negatives) represents instances where the model fails to identify two papers actually authored by the same person. In the reported results, we average the corresponding metrics across all names to obtain a macro-level indicator.

5.1.4. Implementation details

We construct a graph for each ambiguous name and perform transductive learning on the graph. The initial graph is built using three types of relationships: co-author, co-organization, and co-venue. The threshold for co-author is explored from {1, 2, 3}, the threshold for co-organization is explored between 0 and 0.6, and the threshold for co-venue is explored from {1, 2, 3}. The output dimension of Word2Vec is set to 100. The output dimension of the MLP_m and MLP_c projection heads are set to 128. The GAT-v encoder has two hidden layers, each with a dimension of 256, and the parameter t for the proximity matrix is explored from {1, 2, 3}. The hyperparameters *epsilon* and *min_samples* for HDBSCAN are explored from {0.1, 0.15, 0.2} and {2, 3, 4, 5, 6}, respectively. We generate different graph augmentation views based on the degree centrality method in GCA, requiring the parameters feature drop rate $r_{f,1}, r_{f,2}$ and edge drop rate $r_{e,1}, r_{e,2}$ to be selected from the range 0.1 to 0.6 increments of 0.1. We note that the model is not sensitive to the temperature coefficient τ_1, so we set it to 0.3. τ_2 is explored between 0.2 and 0.6 with an interval of 0.1, and the loss weight factor ω is explored from 0.1 to 1.0 with an interval of 0.1. We implement the code using PyTorch and optimize the model using the Adam optimizer (Kingma, 2014). The learning rate is explored between 0.0001 and 0.001. The number of training epochs for each model related to an ambiguous name is set to 50. Hyperparameters are tuned using grid search on the validation set to find the best combination. All experiments are conducted on an NVIDIA GTX 3080 GPU.

Table 2
Disambiguation performance on AMiner-AND dataset (%).

AMiner-AND			
Model	Pre	Rec	F1
AGAND	69.11	56.78	60.14
AMiner	77.96	63.03	67.79
PHNet	61.35	**79.75**	67.33
LAND	78.36	59.68	63.36
MRAND	75.60	67.10	69.60
SND-all	73.14	72.50	68.50
BOND	79.70	67.73	69.67
EAND	79.11	63.68	70.56
MCCG	**81.18**	68.65	**71.24**

Table 3
Disambiguation performance on WhoisWho-v1 dataset (%).

WhoisWho-v1			
Model	Pre	Rec	F1
AGAND	67.97	52.89	57.64
AMiner	81.01	60.78	69.45
PHNet	46.50	55.13	48.69
DHGN	68.34	68.72	62.42
SND-all	76.69	81.96	76.57
BOND	83.75	89.67	84.77
EAND	78.12	77.98	78.05
MCCG	**87.02**	**89.96**	**87.46**

5.2. Main results

The results of our model, MCCG, along with other baselines on four datasets, are shown in Tables 2–5. All experiments were run 5 times, and the average results are reported.

Compared to the other baselines, MCCG achieves the highest improvement in Macro-F1 scores, reaching 11.10 %, 38.77 %, 43.64 %,

Table 4
Disambiguation performance on WhoisWho-v2 dataset (%).

WhoisWho-v2			
Model	Pre	Rec	F1
AGAND	76.12	52.99	61.08
AMiner	84.33	61.97	71.44
PHNet	43.90	60.65	48.93
SND-all	84.90	87.38	85.25
BOND	90.22	93.19	90.89
EAND	79.98	85.61	82.70
MCCG	**92.60**	**93.45**	**92.57**

Table 5
Disambiguation performance on LAGOS-AND dataset (%).

LAGOS-AND			
Model	Pre	Rec	F1
AGAND	65.69	81.03	70.76
AMiner	83.67	82.40	82.26
PHNet	78.22	83.14	81.45
SND-all	77.99	80.13	76.79
BOND	86.05	77.80	79.62
EAND	79.51	84.80	82.07
MCCG	**90.52**	**85.71**	**86.12**

and 15.36 % on the AMiner-AND, WhoisWho-v1, WhoisWho-v2, and LAGOS-AND datasets, respectively. When compared with the latest methods (i.e., SND-all, BOND, and EAND), MCCG exhibits an improvement of 2.74 %, 1.57 %, and 0.68 % in Macro-F1 scores on the AMiner-AND dataset; 10.89 %, 2.69 %, and 9.41 % on the WhoisWho-v1 dataset; 7.32 %, 1.68 %, and 9.87 % on the WhoisWho-v2 dataset; and 9.33 %, 6.5 %, and 4.05 % on the LAGOS-AND dataset, respectively. These results demonstrate the effectiveness of MCCG in addressing the AND task.

A detailed analysis of the disambiguation performance of various models was conducted. AGAND performs poorly across all datasets as it relies solely on relational data without integrating semantic information, which results in embeddings with limited informative content. AMiner shows relatively good performance on datasets with fewer ambiguous papers (e.g., AMiner-AND and LAGOS-AND); however, its performance deteriorates significantly on the more ambiguous WhoisWho-v1 and WhoisWho-v2 datasets, with Macro-F1 scores 18.41 % and 21.13 % lower than those of MCCG, respectively. This is likely because it decouples global metric learning and local linkage learning into two separate processes, resulting in inconsistent embeddings and suboptimal performance when encountering highly ambiguous names. PHNet exhibits a similar trend, performing even worse than AMiner on the WhoisWho-v1 and WhoisWho-v2 datasets. When the total number of papers associated with ambiguous names is very large, the heterogeneous graph convolutional network embedding method (HGCN) proposed by PHNet becomes insufficient. Although it employs random walks to capture neighbor relationship patterns for training HGCN, which works well on small-scale heterogeneous graphs, the complex and diverse neighbor relationships in large-scale graphs tend to obscure the distinctive features of papers. Although LAND leverages multimodal textual information from knowledge graphs to enhance embedding quality, integrating such information does not necessarily produce positive outcomes. For instance, on the AMiner-AND dataset, LAND's Macro-F1 score is 7.88 % lower than that of MCCG, indicating suboptimal disambiguation capability. MRAND, which utilizes an attention-based, multi-dimensional, multi-hop graph convolutional network for representation learning to capture rich neighborhood relationships, achieves a Macro-F1 score on the AMiner-AND dataset that is only 1.64 % lower than MCCG. However, decoupling representation learning from clustering disambiguation fails to deliver optimal overall performance. Both DHGN and SND-all utilize semantic and relational features to encode paper embeddings. DHGN employs FastText for semantic encoding, trained on an extensive corpus from AMiner and DBLP (approximately 1.6 million papers), and uses Hin2Vec to learn relational information from the heterogeneous graph. In contrast, SND-all uses Word2Vec for semantic encoding, drawing training data from the entire dataset without additional information, and employs metapath2vec for relational feature encoding. Ultimately, on the WhoisWho-v1 dataset, DHGN's Macro-F1 is 14.15 % lower than that of SND-all, suggesting that an excessively large training corpus does not necessarily yield positive gains; furthermore, inherent differences between the encoding capacities of metapath2vec and Hin2Vec may also contribute. Overall, the performance of SND-all across the four datasets is less competitive, with Macro-F1 scores that are 3.19 %, 10.89 %, 7.32 %, and 9.33 % lower than those of MCCG, respectively. These disambiguation challenges largely arise from the decoupling of metric learning and clustering processes, as well as the complex neighborhood relationships inherent in highly ambiguous heterogeneous graphs. BOND unifies the metric learning and clustering disambiguation stages to enable mutual reinforcement, achieving better results, its Macro-F1 scores are 1.57 %, 2.69 %, 1.68 %, and 6.5 % lower than those of MCCG on the four datasets, respectively. However, since BOND neither addresses the noise present in the initial graph nor adopts a dynamic cluster label learning approach, its performance remains slightly inferior to that of MCCG. EAND constructs six similarity graphs using paper information and employs a pre-pruning method to improve graph structure, followed by a mixed attention mechanism to extract information from these graphs. This approach enables EAND to achieve competitive performance on the AMiner-AND and LAGOS-AND datasets, with Macro-F1 scores only 0.68 % and 4.05 % lower than those of MCCG. However, on the WhoisWho-v1 and WhoisWho-v2 datasets, EAND's Macro-F1 scores are 9.41 % and 9.87 % lower than MCCG's, respectively. This suggests that although the six similarity graphs provide rich relational information, missing data in certain cases can mpede the construction of some graphs, thereby affecting the overall disambiguation performance.

Additionally, we observe that on the WhoisWho-v1, WhoisWho-v2, and LAGOS-AND datasets, MCCG consistently achieves higher Macro-Precision and Macro-Recall scores than the other baselines, with these two metrics being well balanced. In contrast, on the AMiner-AND dataset, a significant disparity exists between these scores; notably, except for PHNet and SND-all, which achieve relatively high recall, all other baselines exhibit this phenomenon. Further analysis of the AMiner-AND dataset and the experimental results reveals that many papers suffer from missing information, particularly in keywords; for some ambiguous names, as many as 20 % to 45 % of the papers lack keyword data, inevitably affecting their representations. However, PHNet and SND-all manage to achieve higher recall, possibly because they capture relational features via random walks on heterogeneous graphs. This approach allows them to capture more distant neighborhood relationships than graph encoders based on homogeneous graphs, thereby better compensating for the impact of missing semantic information. Nonetheless, this strategy has limitations when dealing with ambiguous names associated with a larger number of papers. Moreover, to investigate other factors that might influence MCCG's recall, we adjusted certain hyperparameters and observed the resulting changes. We found that a slight increase in the clustering radius parameter (*epsilon*) for HDBSCAN yields a better balance between precision and recall. For example, when other parameters are held constant and *epsilon* is set to 0.18, the Macro-Precision, Macro-Recall, and Macro-F1 scores reach 76.37 %, 72.12 %, and 71.96 %, respectively. This improvement is attributed to a larger clustering radius incorporating more edge-case samples into clusters, thereby enhancing recall. However, we deem overly fine-grained hyperparameter tuning unnecessary; therefore, in our experiments, *epsilon* is chosen from the set {0.1, 0.15, 0.2}.

To demonstrate the effect of MCCG in more detail, we selected the disambiguation results of 15 highly ambiguous names from each of the four datasets and compared them with other baselines. The results are presented in Tables A.9–A.12.

Table 6
Ablation study on multi-view contrastive module (%).

Model	AMiner-AND			WhoisWho-v1			WhoisWho-v2			LAGOS-AND		
	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1
MCCG-w/o MC	**81.34**	67.42	70.59	85.85	**90.25**	86.90	91.46	93.36	91.67	**90.83**	84.72	85.66
MCCG-TCL	81.15	66.07	69.39	85.99	90.24	86.96	91.73	93.44	92.12	90.03	84.85	85.26
MCCG	81.18	**68.65**	**71.24**	**87.02**	89.96	**87.46**	**92.60**	**93.45**	**92.57**	90.52	**85.71**	**86.12**

5.3. Effect of multi-view contrastive module

5.3.1. Ablation analysis

In order to evaluate the effectiveness of the multi-view contrastive module, we develop two variants of MCCG and evaluate them on four datasets, with the results presented in Table 6. In one variant, the multi-view contrastive module is removed (denoted as MCCG-w/o MC), and in the other variant, pseudo-label cluster guidance signals are omitted and the contrastive loss is replaced by the traditional contrastive loss (van den et al., 2018) (denoted as MCCG-TCL).

As shown in the experimental results in Table 6, the Macro-F1 performance of MCCG-w/o MC decreases by 0.65 %, 0.56 %, 0.9 %, and 0.46 % on the four datasets, respectively, compared to MCCG. This indicates that the multi-view contrastive module contributes to improving the model's performance. In comparison, MCCG-TCL experiences a decrease in Macro-F1 by 1.85 %, 0.5 %, 0.45 %, and 0.86 % on the four datasets, respectively. Specifically, MCCG-TCL outperforms MCCG-w/o MC by 0.06 % and 0.45 % on the WhoisWho-v1 and WhoisWho-v2 datasets, but underperforms by 1.2 % and 0.4 % on the AMiner-AND and LAGOS-AND datasets, respectively. These results suggest that MCCG-TCL offers some benefits on the WhoisWho-v1 and WhoisWho-v2 datasets, but has a negative impact on the AMiner-AND and LAGOS-AND datasets. In contrast, our proposed cross-view CL method yields positive gains across all four datasets and demonstrates more stable performance compared to traditional contrastive methods. This suggests that the cross-view CL method is better suited to the practical requirements of the AND task, as it more effectively extracts detailed information at both the view and cluster levels, thereby helping the model better mitigate noise interference.

5.3.2. Sensitivity analysis of drop rate

In this section, we analyze the feature drop and edge drop rate parameters during graph augmentation, including $r_{f,1}, r_{f,2}, r_{e,1}$, and $r_{e,2}$, to explore how varying levels of feature and edge drop, i.e., the proportion of noise introduced, affect the robustness of the model. We conduct experiments on the WhoisWho-v1 dataset. For visualization convenience, we set $r_f = r_{f,1} = r_{f,2}$ and $r_e = r_{e,1} = r_{e,2}$, so that the noise introduction is controlled by r_f and r_e, the range is from 0.1 to 0.6. The experimental results are shown in Fig. 4.

In the experimental results, it is observed that when minimal noise is introduced, that is, when r_f and r_e are set to small values, the model exhibits poor disambiguation performance, indicating a lack of robustness against the inherent noise in the original graph. As r_f and r_e gradually increase and reach an optimal level, the performance improves, suggesting that a moderate amount of noise can help enhance the model's robustness against interference. However, when r_f and r_e continue to increase beyond this optimal point, performance eventually declines, revealing that excessive noise exerts a detrimental effect. Thus, these

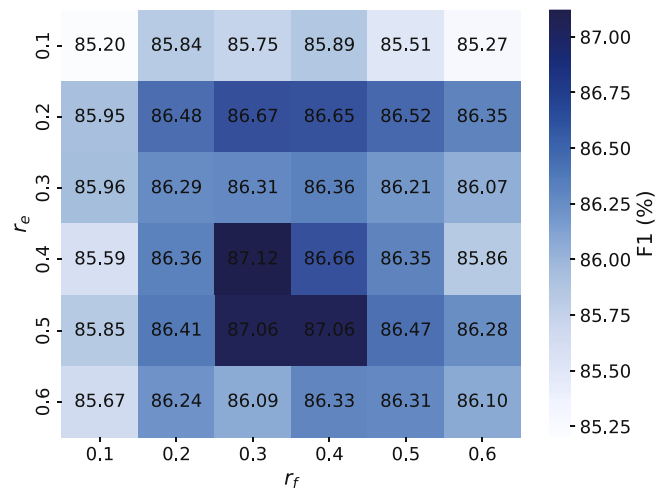


Fig. 4. Sensitivity analysis of drop rate.

findings further validate the effectiveness of the multi-view contrastive module in enhancing the model's robustness.

5.4. Effect of cluster-guided module

5.4.1. Ablation analysis

To further investigate the effectiveness of the cluster-guided module, we develop two variants for comparison with MCCG. In one variant, denoted as MCCG-w/o CG, the cluster-guided contrastive loss is removed, and the embeddings output by the MLP_c projection head are directly fed into HDBSCAN to generate pseudo-labels for guiding model training. In the other variant, denoted as MCCG-w/o AW, the cluster-guided contrastive loss is retained but adaptive weighting (AW) is not applied, so that each sample pair is assigned the same weight.

The results in Table 7 show that MCCG-w/o CG experiences a significant decrease in Macro-F1 by 1.91 %, 9.38 %, 12.39 %, and 3.89 % across the four datasets compared to MCCG. This consistent and significant performance drop indicates that, without the guidance from the cluster-guided module, model training tends to diverge uncontrollably. Furthermore, removing the AW method leads to performance declines of 2.48 %, 1.39 %, 1.26 %, and 1.94 % compared to MCCG, highlighting its importance. The substantial performance loss suggests that, without the ability to dynamically compute sample pair weights, the model struggles with ambiguous decision boundaries for hard samples, making it harder to correct misclassified samples over multiple training iterations. However, by incorporating the AW method, MCCG effectively addresses this issue, enabling the model to dynamically refine its embeddings and

Table 7
Ablation study on cluster-guided module (%).

Model	AMiner-AND			WhoisWho-v1			WhoisWho-v2			LAGOS-AND		
	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1
MCCG-w/o CG	75.48	**71.33**	69.33	73.87	**93.47**	78.08	72.79	**97.70**	80.18	84.73	82.52	82.23
MCCG-w/o AW	**86.35**	62.08	68.76	84.09	91.77	86.07	89.98	94.24	91.31	83.35	**86.00**	84.18
MCCG	81.18	68.65	**71.24**	**87.02**	89.96	**87.46**	**92.60**	93.45	**92.57**	**90.52**	85.71	**86.12**

enhance the accuracy of pseudo-label guidance. The improved representation quality further drives the guidance signals in the right direction, and, in turn, more accurate guidance signals reinforce representation learning, creating a positive feedback loop.

5.4.2. Reliability analysis of cluster-guided signals

The reliability of clustering guidance pseudo-labels is crucial, as they affect the overall training process of the model. In our work, we propose an adaptive sample weighting method (AW) for clustering guidance, which enables the model to dynamically learn sample pair weights in the contrastive loss during training. This enables the model to focus more on hard and false samples, thereby preventing the accumulation of erroneous guidance signals. To assess the reliability of the guidance signals and the effectiveness of our proposed method, we conducted additional experiments comparing MCCG and MCCG-w/o AW. We evaluate the methods from two perspectives: local sampling and global evaluation. For the local sampling experiment, we selected two highly ambiguous names from the WhoisWho-v1 dataset: "weimin_liu" and "feng_wang" (detailed information is provided in Table A.8). Every ten epochs during training, we compared the pseudo-labels with the ground truth labels. Since the predicted and true labels are matched on a pairwise basis rather than one-to-one, we adopt the *Jaccard Index* to evaluate the accuracy of the pseudo-labels, which is calculated as $\frac{|TP|}{|TP|+|FP|+|FN|}$. The value ranges from 0 to 1, with higher values corresponding to better label prediction accuracy and, consequently, more reliable clustering guidance signals. The experimental results are shown in Fig. 5(a). For the global evaluation experiment, both models were assessed every ten epochs on the WhoisWho-v1 dataset in terms of Macro-Precision, Macro-Recall, and Macro-F1. The results are shown in Fig. 5(b)–(d).

In Fig. 5(a), the red curve represents the results for "weimin_liu" and the green curve those for "feng_wang". The solid and dashed lines correspond to the MCCG and MCCG-w/o AW models, respectively. The results show that in the early stages of training, both models primarily rely on the information provided by semantic features for clustering, without fully integrating relational features. As a result, the accuracy of the pseudo-labels is limited, and the *Jaccard Index* is relatively low. However, as training progresses, both models show significant improvements. Ultimately, MCCG outperforms the MCCG-w/o AW model in terms of the *Jaccard Index*. This indicates that our proposed AW method has a significant effect on improving the accuracy of pseudo-labels. When faced with samples with ambiguous clustering boundaries, such as those with similar research fields but different co-authors, MCCG-w/o AW struggles to detect these subtle differences. In contrast, MCCG, by dynamically learning sample weights, is better at distinguishing these hard-to-classify samples. Specifically, we note that "weimin_liu" exhibited a decline in performance at epoch 30 relative to epoch 20, but improved markedly by epoch 40. This is likely because the model overemphasized hard samples around epoch 30, assigning them excessively high weights at the expense of overall clustering consistency. At this stage, the model had not yet converged; however, as it further adjusted the weights, stable clustering performance was achieved after epoch 40. This suggests that the adaptive sample weighting method played a key role in correcting erroneous guidance signals, preventing the accumulation of errors.

The results in Fig. 5(b)–(d) show that as the number of training epochs increases, both MCCG and MCCG-w/o AW exhibit remarkable improvements in Macro-Precision and Macro-F1, while Macro-Recall shows a noticeable decrease. This suggests that the models sacrifice some recall to achieve higher precision during training. However, MCCG demonstrates a larger overall gain in Macro-Precision compared to MCCG-w/o AW, with the former achieving an approximately 5 % increase and the latter about 4 %. Additionally, the overall loss in Macro-Recall is smaller for MCCG, with a decrease of about 2 %, while MCCG-w/o AW shows a decline of about 4 %. These results indicate that MCCG is more confident when facing ambiguous, hard samples, and that the AW method allows the model to detect more complex patterns between papers, preventing overfitting to simpler samples, alleviating the rapid decline in recall, and preserving generalization to difficult samples. In contrast, the MCCG-w/o AW model is more prone to overfitting high-frequency patterns, sacrificing recall for long-tail samples. Furthermore, a brief drop in Macro-F1 for MCCG is observed at epoch 30. With the help of AW method, the model escapes from a local optimum and achieves a more impressive result in the final outcome.

5.5. Embedding analysis

To better visualize the spatial distribution of the embeddings generated by our proposed variants, we employ t-SNE (van der Maaten & Hinton, 2008) to project the embeddings onto a 2-dimensional space for comparative analysis. We selected the highly ambiguous name "jie_wu" from the WhoisWho-v1 dataset, which contains 84 authors sharing this name and comprises a total of 2590 papers. The corresponding visualization results are shown in Fig. 6.

We primarily analyze the blue points in Fig. 6. It can be observed that in Fig. 6(d) the blue points exhibit a higher degree of concentration and more distinct decision boundaries from other colored points. Furthermore, Fig. 6(a) shows a relatively blurred decision boundary and performs poorly across all three evaluation metrics. Subsequently, Fig. 6(b) reveals that the clusters are more dispersed, accompanied by a sharp decline in the pairwise precision score for disambiguation. Finally, Fig. 6(c) also yields unsatisfactory disambiguation outcomes.

These results demonstrate that the multi-view contrastive module effectively mitigates the interference caused by noise and is capable of extracting more fine-grained features. The cluster-guided module plays a crucial role in the overall AND task, as it not only provides high-confidence guidance signals but also generates discriminative embeddings. Additionally, the adaptive sample weighting method effectively alleviates the accumulation of errors arising from incorrect labels during the training process.

5.6. Parameters analysis

To analyze the impact of hyperparameters on AND, we examined the following: the HDBSCAN hyperparameters, *epsilon* and *min_samples*;

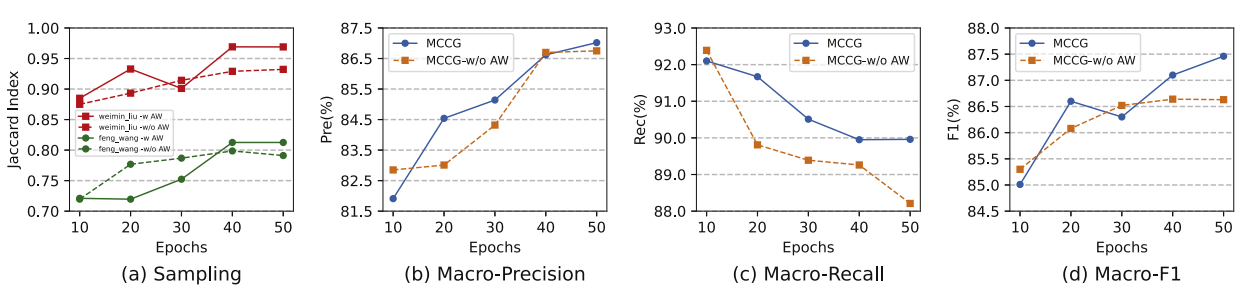
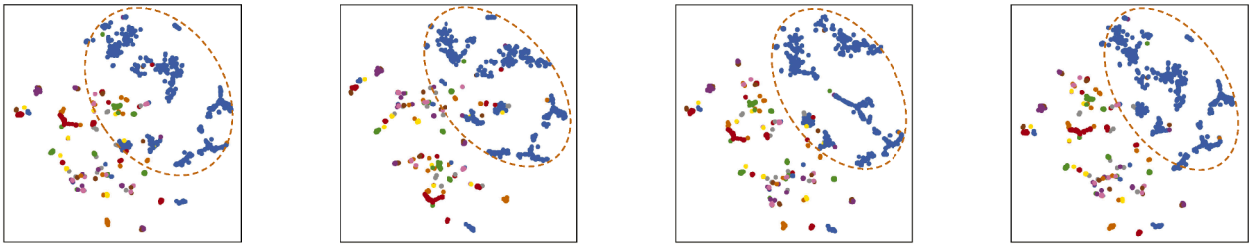


Fig. 5. Reliability analysis of cluster-guided signal.



(a) MCCG-w/o MC 87.22 / 88.77 / 87.99 (b) MCCG-w/o CG 44.24 / 99.15 / 61.18 (c) MCCG-w/o AW 48.47 / 92.75 / 63.66 (d) MCCG 88.23 / 90.11 / 89.16

Fig. 6. The t-SNE visualization of name "jie_wu" in WhoisWho-v1 dataset. Each dot represents a paper, with dots of the same color indicating papers by the same author in the ground truth. Panels (a), (b), (c) respectively denotes the results of different variants of (d). The second row of metrics corresponds to Pre, Rec, and F1 (%).

the loss weight factor ω; the temperature coefficient τ_2; the proximity matrix parameter t in GAT-v.

5.6.1. Parameters of HDBSCAN

In this work, we chose HDBSCAN as the clustering algorithm and further conducted a sensitivity analysis on its hyperparameters. Both the maximum distance between neighboring samples (*epsilon*) and the minimum number of samples within a neighborhood (*min_samples*) impact the final performance. Fig. 7 shows that the value of *epsilon* exerts a significant influence on clustering performance. This effect is especially pronounced on the AMiner-AND, WhoisWho-v1, and WhoisWho-v2 datasets. Moreover, on the LAGOS-AND dataset, when *min_samples* is held constant, variations in *epsilon* yield more consistent performance trends, indicating that careful tuning of *epsilon* can lead to improved clustering outcomes. In addition, the value of *min_samples* itself significantly impacts performance. Relatively small *min_samples* help prevent small clusters from being mistakenly identified as noise, thereby enabling a more accurate detection of diverse clustering structures. Conversely, setting *min_samples* too high may cause HDBSCAN to fail to form clusters in low-density regions, which in turn hinders the recognition of authors with fewer papers.

5.6.2. Sensitivity of weight factor

During model training, ω serves as the trade-off parameter between $\mathcal{L}_m$ and $\mathcal{L}_c$. We explored its optimal value within the range of 0.1 to 1.0 in increments of 0.1. As shown in Fig. 8(a), the model performance tends to be optimal when ω is set to 0.5 or 0.6. This indicates that the multi-view contrastive module and the cluster-guided module jointly enhance the disambiguation performance.

5.6.3. Sensitivity of temperature coefficient

We also conducted a sensitivity analysis on the temperature coefficients. In our experiments, we observed that the model is not sensitive to τ_1; therefore, based on empirical observations, we fixed it at 0.3. We further analyzed the sensitivity of τ_2 by exploring values from 0.2 to 0.6 in increments of 0.1. The results in Fig. 8(b) show that for the

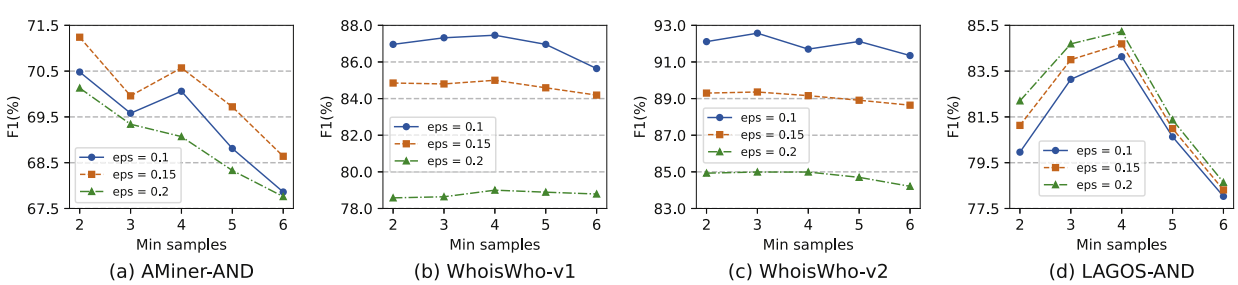
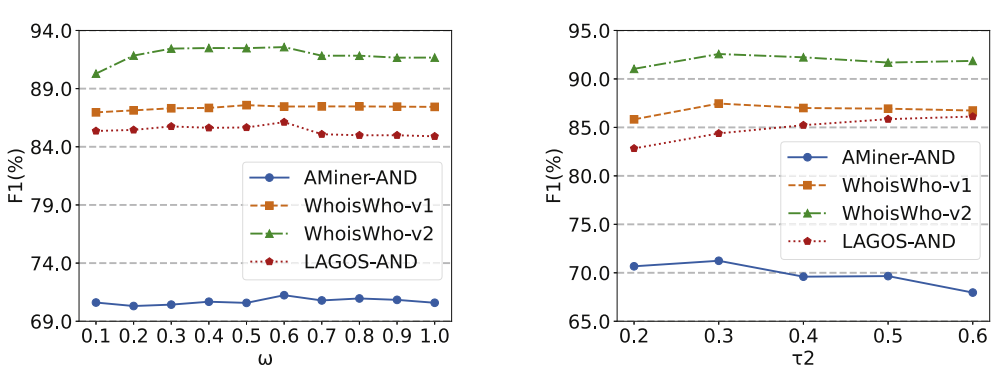


Fig. 7. Parameters analysis of HDBSCAN.



(a) Loss weight factor ω (b) Temperature coefficient τ_2

Fig. 8. Parameters sensitivity analysis of ω and τ_2.

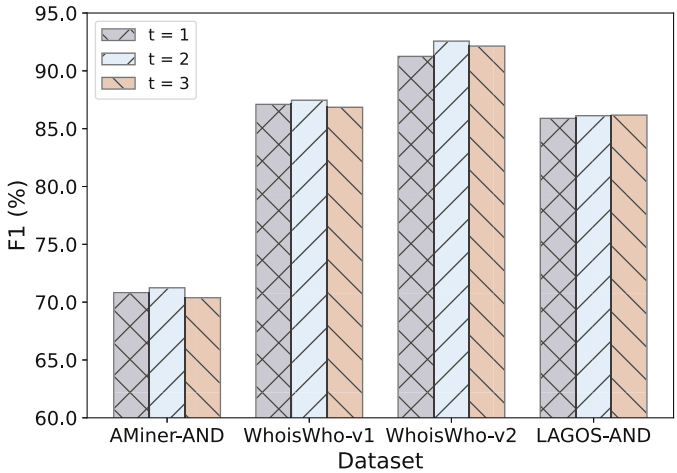


Fig. 9. Parameter analysis of t in GAT-v.

AMiner-AND, WhoisWho-v1, and WhoisWho-v2 datasets, the best performance is achieved at $\tau_2 = 0.3$, beyond which the F1 score gradually declines. In contrast, on the LAGOS-AND dataset, the optimal performance is obtained when $\tau_2 = 0.6$. These findings indicate that selecting an appropriate temperature coefficient can, to a certain extent, enhance disambiguation performance.

5.6.4. Sensitivity of the t in GAT-v

In our work, we adopt a variant of the Graph Attention Network (GAT), termed GAT-v (Wang et al., 2019). Unlike the original GAT, GAT-v introduces a tunable parameter t that allows the model to capture neighbor information at different orders, thereby obtaining richer relational cues for disambiguation. We conducted parameter experiments on four datasets, and the corresponding results are presented in Fig. 9.

The results indicate that for the AMiner-AND, WhoisWho-v1, and WhoisWho-v2 datasets, MCCG achieves optimal performance at $t = 2$, while on the LAGOS-AND dataset a slight improvement is observed at $t = 3$. In contrast, when $t = 3$ the performance of the first three datasets declines slightly, and at $t = 1$the performance across all datasets is inferior compared to $t = 2$. These findings suggest that limiting the model to first-order neighbors prevents it from fully exploiting potentially useful information. Appropriately aggregating higher-order neighbor information enriches the latent features in the embeddings, which is particularly beneficial for nodes in sparse regions of the graph where first-order neighbors alone may not provide clear discriminative cues. Nonetheless, incorporating overly distant neighbors may introduce ambiguous data that adversely affects the accurate representation of a node's own features.

6. Conclusion

In this work, we propose a unified end-to-end framework for author name disambiguation (AND) named MCCG. On one hand, MCCG enhances the model's robustness to noise in the graph; on the other hand, it leverages the clustering effect of homogeneous papers to integrate the representation learning and clustering-based disambiguation stages. This integration enables a contrastive learning method that can correct clustering outcomes. Specifically, we introduce controlled noise into the graph and employ a multi-view contrastive module to perform contrastive learning at both the view and cluster levels, thereby further enhancing the model's robustness. Experimental results demonstrate that our approach significantly enhances model robustness. Additionally, we unify representation learning and clustering tasks through the clustering effect, which promotes the learning of discriminative embeddings and generating high-confidence clustering guidance signals that mutually reinforce each other. Moreover, the self-correcting capability of the cluster labels underscores the superiority of our model. Experiments on four datasets show that MCCG outperforms existing state-of-the-art models.

7. Future work

For literature management systems, author name disambiguation (AND) is a complex and large-scale task that cannot realistically rely solely on human effort. There is a clear trend toward employing intelligent technologies for disambiguation. Therefore, improving disambiguation accuracy is crucial, as it impacts various aspects such as system management and user experience. In our experiments, we observed that when paper attribute information is severely lacking, the limited semantic and relational data available for generating paper embeddings fails to provide clear decision boundaries, resulting in suboptimal disambiguation performance. Building on this issue, we plan to develop our future work from two perspectives that may enhance disambiguation outcomes. First, we plan to explore an Incorrect Assignment Detection (Chen et al., 2023) task, which aims to re-examine the already determined paper affiliations, identify potential misclassifications, and correct them, thereby further improving disambiguation performance. Second, we are inspired by the impressive performance of large language models (LLMs) across various domains. Given their extensive knowledge bases and access to vast amounts of paper-related data, fine-tuning these models to leverage their understanding and inference capabilities may yield surprising improvements in disambiguation results.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Fan Ye: Writing – review & editing, Validation, Supervision, Funding acquisition, Resources, Project administration; **Zong Xia:** Writing – original draft, Writing – review & editing, Methodology, Formal analysis, Data curation; **Zhaolong Ling:** Writing – review & editing, Supervision, Conceptualization, Funding acquisition; **Le Wu:** Writing – review & editing, Supervision, Conceptualization.

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Appendix A. Sampling experiments

To more intuitively demonstrate the disambiguation performance of MCCG, we selected 15 highly ambiguous names from each of the four datasets based on the number of authors and papers, and compared the results with those of other baselines. The statistical information of the selected ambiguous names is presented in Table A.8. The final disambiguation performance of each model is shown in Tables A.9–A.12.

Table A.8
Statistical information of the selected ambiguous names.

AMiner-AND			WhoisWho-v1			WhoisWho-v2			LAGOS-AND		
name	# authors	# papers	name	# authors	# papers	name	# authors	# papers	name	# authors	# papers
lei_song	34	700	gang_li	157	3228	xiang_li	95	2117	wei_wang	36	356
lin_huang	31	553	rui_wang	143	2643	qiang_wang	76	2281	wei_li	16	228
yang_shen	30	485	tao_li	138	2161	ping_chen	58	1710	jie_zhang	15	221
ping_sun	30	524	jian_yang	120	2863	xiaofeng_wang	53	1392	xin_wang	14	210
song_chen	28	598	jie_gao	102	1254	min_liu	51	1292	bin_zhang	13	117
xu_xu	28	555	bin_zhou	95	1529	ping_wu	37	1261	qiang_li	13	149
lu_han	24	366	jie_wu	84	2590	weihua_wang	31	755	yan_wang	11	255
c_y_huang	24	203	fei_chen	79	1412	lingling_wang	27	827	lei_chen	10	237
yong_tian	16	262	yao_wang	77	1406	qing_jiang	25	959	hui_zhang	10	109
shuai_yuan	16	310	bing_xu	68	1236	makoto_inoue	18	634	michael_smith	9	147
hong_fan	16	366	jie_tang	43	1198	xiaolin_wu	14	400	yong_zhang	9	184
dan_sun	16	204	weimin_liu	30	1434	feng_peng	14	375	dong_wang	9	193
tao_deng	15	306	feng_wang	19	741	wendong_wang	12	480	stephen_smith	9	208
hongbin_li	15	286	biao_huang	14	784	ying_tan	11	316	yi_li	8	169
hua_fu	12	415	hongjie_zhang	12	626	xi_yao	8	881	dinesh_kumar	8	127

Table A.9
Detailed results on AMiner-AND dataset (%).

Name	MCCG			BOND			SND-all			AMiner			AGAND		
	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1
lei_song	59.21	88.98	71.10	62.12	88.45	**72.98**	53.49	83.77	65.29	61.99	73.09	67.09	53.40	88.95	66.74
lin_huang	81.32	58.55	**68.08**	78.44	52.75	63.08	69.25	62.42	65.66	73.86	31.52	44.18	69.59	32.61	44.41
yang_shen	79.08	52.57	**63.15**	79.13	51.32	62.26	69.94	51.20	59.12	74.62	47.38	57.96	49.29	42.83	45.83
ping_sun	64.92	68.30	**66.57**	63.54	61.92	62.72	46.08	79.05	58.22	63.72	53.58	58.21	41.29	45.07	43.10
song_chen	40.98	33.35	36.77	55.13	30.03	38.88	45.97	33.80	**38.96**	44.77	29.75	35.75	39.35	22.38	28.54
xu_xu	64.09	63.52	63.80	71.36	56.26	62.92	69.49	66.03	**67.72**	71.14	58.93	64.46	47.93	41.21	44.32
lu_han	47.73	45.40	**46.53**	49.23	26.85	34.75	44.72	26.03	32.91	59.50	30.25	40.11	41.26	19.37	26.36
c_y_huang	43.02	86.81	57.53	33.25	73.40	45.77	15.37	87.23	26.14	51.05	70.11	**59.08**	34.89	52.55	41.94
yong_tian	84.42	61.31	**71.04**	74.93	57.93	65.34	54.18	62.29	57.95	81.07	55.25	65.71	70.75	55.28	62.07
shuai_yuan	84.03	80.64	**82.30**	93.08	50.41	65.40	86.46	52.75	65.53	47.46	39.71	43.24	72.74	51.13	60.05
hong_fan	92.54	84.75	**88.47**	91.82	82.41	86.86	55.78	85.46	67.50	76.88	67.32	71.78	74.43	72.19	73.29
dan_sun	83.64	90.74	**87.05**	87.05	86.76	86.91	34.77	88.15	49.87	79.98	82.59	81.27	73.23	62.83	67.63
tao_deng	72.34	54.74	**62.32**	79.67	41.95	54.96	60.28	43.90	50.80	76.62	43.42	55.43	52.73	28.63	37.11
hongbin_li	89.90	75.61	**82.14**	79.17	71.75	75.28	50.64	80.01	62.02	65.96	70.19	68.01	54.13	50.80	52.41
hua_fu	68.55	96.89	**80.29**	68.63	95.17	79.75	67.49	94.56	78.77	81.80	77.77	79.73	57.81	71.49	63.93

Table A.10
Detailed results on WhoisWho-v1 dataset (%).

Name	MCCG			BOND			SND-all			AMiner			AGAND		
	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1
gang_li	63.19	85.63	72.72	63.32	86.05	**72.96**	43.57	78.85	56.13	66.94	36.74	47.44	38.32	51.03	43.77
rui_wang	60.15	81.51	**69.22**	58.67	79.83	67.64	35.74	69.32	47.16	63.38	41.91	50.46	37.65	41.46	39.47
tao_li	52.82	84.43	**64.98**	43.81	90.60	59.06	24.92	72.45	37.08	46.68	45.18	45.92	36.66	57.50	44.77
jian_yang	68.92	92.71	**79.06**	35.90	91.60	51.59	52.90	83.17	64.67	73.64	35.58	47.98	61.29	61.16	61.23
jie_gao	67.28	95.07	**78.80**	65.31	94.61	77.28	44.42	86.71	58.75	53.12	42.88	47.45	36.91	49.42	42.26
bin_zhou	67.61	97.85	**79.97**	63.80	97.71	77.19	52.63	89.54	66.29	80.32	38.58	52.12	70.09	65.63	67.79
jie_wu	88.23	90.11	89.16	49.03	96.48	65.02	93.98	89.28	**91.57**	87.27	24.95	38.80	77.39	42.69	55.02
fei_chen	90.06	97.28	93.53	94.19	97.23	**95.69**	72.05	89.62	79.88	85.81	65.07	74.02	56.75	55.26	56.00
yao_wang	67.21	72.83	**69.91**	58.10	78.52	66.78	53.24	75.38	62.40	60.65	31.60	41.55	39.66	29.70	33.97
bing_xu	74.21	96.31	**83.83**	71.19	94.87	81.34	56.26	81.23	66.48	73.40	40.67	52.34	46.75	54.81	50.46
jie_tang	92.93	90.87	**91.89**	83.96	90.74	87.21	89.32	80.33	84.59	91.90	63.01	74.76	74.02	64.64	69.02
weimin_liu	99.33	97.54	98.43	99.34	98.77	**99.05**	98.27	83.58	90.33	94.51	22.76	36.69	97.82	63.14	76.74
feng_wang	90.40	88.92	**89.65**	89.49	88.24	88.86	94.19	82.85	88.16	75.45	72.70	74.05	58.26	51.18	54.49
biao_huang	83.82	99.56	**91.02**	83.79	98.12	90.39	70.49	34.20	46.06	85.17	61.77	71.61	70.49	34.20	46.06
hongjie_zhang	99.46	64.94	**78.58**	99.79	63.65	77.73	96.65	52.22	67.80	96.85	59.96	74.06	95.96	48.44	64.38

Table A.11
Detailed results on WhoisWho-v2 dataset (%).

Name	MCCG			BOND			SND-all			AMiner			AGAND		
	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1
xiang_li	80.05	96.54	**87.53**	78.64	95.83	86.38	73.71	91.87	81.79	71.24	46.85	56.53	52.95	53.14	53.05
qiang_wang	93.93	88.36	**91.06**	92.94	87.42	90.10	83.00	81.03	82.00	76.74	41.42	53.80	61.85	57.73	59.72
ping_chen	75.83	74.04	**74.92**	61.63	74.14	67.31	74.77	66.60	70.45	55.58	34.29	42.41	47.22	40.59	43.65
xiaofeng_wang	89.92	96.36	**93.03**	88.45	86.38	87.41	87.98	82.89	85.36	73.18	49.34	58.94	55.41	56.33	55.87
min_liu	72.04	94.72	**81.84**	68.32	95.38	79.61	61.09	88.92	72.43	67.39	63.95	65.62	46.16	63.90	53.60
ping_wu	88.72	85.62	87.15	91.07	96.75	**93.83**	85.54	86.63	86.09	70.85	51.08	59.37	62.65	54.40	58.24
weihua_wang	97.57	89.63	**93.43**	97.22	81.46	88.65	94.99	77.37	85.28	88.65	42.60	57.54	85.65	37.10	51.80
lingling_wang	93.67	99.50	96.50	94.19	99.00	**96.54**	95.15	95.21	95.18	94.68	66.12	77.86	84.48	74.05	78.93
qing_jiang	78.78	92.50	85.09	79.49	96.92	**87.34**	75.39	81.37	78.27	79.94	32.47	46.18	79.80	50.86	62.12
makoto_inoue	87.77	98.01	**92.61**	87.79	97.24	92.28	65.09	81.46	72.36	80.60	94.58	87.03	55.48	61.33	58.26
xiaolin_wu	97.45	99.00	**98.66**	97.34	94.00	95.64	92.75	85.90	89.19	95.81	45.50	61.70	92.71	39.60	55.50
feng_peng	98.18	98.05	**98.12**	67.85	99.14	80.56	68.45	97.83	80.55	92.87	90.11	91.47	82.92	68.38	74.95
wendong_wang	98.00	98.66	**98.33**	98.03	95.86	96.93	99.69	93.25	96.36	94.97	54.47	69.24	94.84	56.11	70.51
ying_tan	94.75	99.71	**97.16**	95.09	95.79	95.44	81.21	95.89	87.94	78.75	59.52	67.79	70.55	49.08	57.88
xi_yao	88.13	99.73	**93.58**	87.90	96.27	91.89	87.74	91.77	89.71	87.85	71.72	78.97	94.87	38.56	54.83

Table A.12
Detailed results on LAGOS-AND dataset (%).

Name	MCCG			BOND			SND-all			AMiner			AGAND		
	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1
wei_li	70.98	75.64	**73.23**	61.03	75.30	67.42	41.80	70.75	52.55	68.02	62.53	65.19	17.69	75.57	28.67
jie_zhang	54.66	47.76	50.98	32.02	38.16	34.82	30.46	73.19	43.01	89.64	59.29	**71.37**	26.99	79.19	40.26
xin_wang	43.85	68.87	53.58	48.85	64.52	55.60	52.44	81.61	63.85	76.36	92.04	**83.47**	13.16	71.98	22.25
bin_zhang	56.81	75.93	**65.00**	35.71	67.48	46.71	20.28	90.89	33.16	46.56	62.66	53.74	15.51	79.51	25.96
qiang_li	80.21	83.07	**81.61**	55.67	72.48	62.97	14.85	86.73	25.35	71.96	87.49	78.97	12.18	81.29	21.19
yan_wang	79.21	39.86	53.03	71.62	37.27	49.03	80.31	50.37	**61.91**	63.58	49.04	55.70	25.60	92.57	40.11
lei_chen	80.65	77.24	**78.91**	63.61	63.39	63.50	64.09	86.10	73.48	62.66	74.80	68.32	28.31	92.38	43.34
hui_zhang	79.67	73.32	**76.36**	42.69	65.64	51.73	23.55	84.78	36.87	67.86	80.68	72.94	14.44	72.63	24.09
michael_smith	92.14	95.46	**93.77**	92.14	92.48	92.31	85.71	86.14	85.93	89.08	94.56	91.74	61.18	92.85	73.76
yong_zhang	94.51	85.32	**89.68**	90.57	76.51	82.95	80.76	81.83	81.29	82.28	81.20	81.72	17.42	75.66	28.32
dong_wang	89.51	82.26	**85.73**	83.16	79.37	81.22	21.40	88.91	34.50	73.78	78.83	76.22	19.37	85.64	31.59
stephen_smith	76.51	95.57	84.99	91.57	92.42	92.00	84.42	88.16	86.25	91.80	95.22	**93.48**	67.72	83.21	74.67
yi_li	95.33	86.02	**90.44**	84.14	74.84	79.22	81.90	69.73	75.33	90.92	59.77	72.12	28.38	54.71	37.37
dinesh_kumar	76.08	68.56	**72.12**	53.02	66.58	59.03	23.98	74.94	36.34	71.25	63.45	67.17	18.47	77.30	29.81

Appendix B. Parameters and components analysis

B.1. Effect of different graph augmentation methods in GCA

GCA provides three graph augmentation methods at the topology level: degree centrality, eigenvector centrality, and PageRank centrality. To compare their impact on disambiguation performance, we conducted a parameter analysis. The experimental results are shown in Table B.13.

The results indicate that across the four datasets, the difference between the best and worst F1 scores is only 0.66 %, 0.7 %, 1.45 %, and 0.08 %, respectively. Clearly, these differences are not significant, suggesting that varying graph augmentation methods do not substantially disrupt the disambiguation outcomes, which is consistent with the findings from our GCA experiments. Therefore, to reduce the computational complexity of parameter tuning, we uniformly employ degree centrality for graph augmentation.

B.2. Sensitivity of graph construction thresholds

We construct the initial graph using three types of relationships: co-author (CoA), co-organization (CoO), and co-venue (CoV). An edge is established between a pair of papers if any one of these relationships exceeds a given threshold. Similar to BOND (Cheng et al., 2024), we compute the similarity for CoA and CoV using *word overlap*, and for CoO using the *Jaccard Index*. The threshold for CoA is explored from the set {1, 2, 3}; for CoO, from 0.2 to 0.6 (in increments of 0.1); and for CoV, from the set {1, 2, 3}. For example, a CoA threshold of "1" means that an edge is created when CoA >= 1, and a CoA + O threshold of "1, 0.2" indicates that an edge is established when either CoA >= 1 or CoO >= 0.2. Table B.14 shows the relevant parameter analysis on the WhoisWho-v1 dataset.

The results show that the co-author relationship has the greatest impact on overall performance, which aligns with our fundamental un-

Table B.13
Performance of different graph augmentation methods (%).

Model	AMiner-AND			WhoisWho-v1			WhoisWho-v2			LAGOS-AND		
	Pre	**Rec**	**F1**	**Pre**	**Rec**	**F1**	**Pre**	**Rec**	**F1**	**Pre**	**Rec**	**F1**
degree	**81.18**	68.65	71.24	**87.02**	89.96	**87.46**	92.60	93.45	92.57	90.52	85.71	86.12
evc	80.97	**69.33**	**71.40**	85.92	**90.26**	86.76	91.45	92.84	91.24	**90.58**	**85.82**	**86.20**
pr	81.08	67.85	70.74	86.80	89.74	87.07	**92.84**	**93.54**	**92.69**	90.55	85.71	86.15

Table B.14
Thresholds of initial graph construction on the WhoisWho-v1 dataset.

Thresholds	CoA	Thresholds	CoA + O	Thresholds	CoA + O + V
1	**78.80**	1, 0.2	82.13	1, 0.5, 1	85.25
2	52.23	1, 0.3	85.76	1, 0.5, 2	**87.46**
3	41.20	1, 0.4	87.28	1, 0.5, 3	86.81
-	-	1, 0.5	**87.33**	-	-
-	-	1, 0.6	86.70	-	-

derstanding. Since ambiguity in co-author names is relatively rare, co-author information serves as a crucial criterion for determining a paper's authorship. Although the impacts of co-organization and co-venue are less pronounced, appropriate threshold adjustments can still yield considerable performance improvements.

B.3. Impact of clustering algorithm

Both major modules of MCCG utilize pseudo labels generated by a clustering algorithm as guidance signals to select positive and negative samples for CL. Moreover, the final performance of the AND task largely depends on the chosen clustering algorithm. Therefore, we explored various options for selecting an appropriate clustering algorithm. HAC (Hierarchical Agglomerative Clustering) requires specifying the final number of clusters; however, in practical scenarios, the true number of authors corresponding to ambiguous names is unknown, rendering HAC unsuitable for literature management systems. In contrast, density-based clustering algorithms, which do not require a predetermined number of clusters, are widely favored by researchers. Algorithms such as DBSCAN (Ester et al., 1996), HDBSCAN (McInnes & Healy, 2017), OPTICS (Ankerst, Breunig, Kriegel, & Sander, 1999), and AP Clustering (Frey & Dueck, 2007) all belong to this category. To compare their effectiveness in the AND task, we conducted experiments on four datasets, with the results presented in Table B.15.

The results show that HDBSCAN achieves the best overall performance, followed by DBSCAN; OPTICS performs similarly to DBSCAN, while AP Clustering performs the worst. This is because HDBSCAN improves upon DBSCAN by addressing its limitations in handling clusters with varying densities and noisy data. Such improvements align well with the characteristics of the AND task, where the number of papers associated with ambiguous names is unevenly distributed, and the density and scale of clusters can vary significantly. HDBSCAN can adaptively capture these variations, leading to better overall performance. For OPTICS, when the *cluster_method* parameter is set to "xi", disambiguation performance deteriorates significantly; however, when it is set to "dbscan", the algorithm adopts the same density connectivity rule as DBSCAN, resulting in comparable performance. AP Clustering relies on an iterative message-passing mechanism between "responsibility" and "availability" to determine cluster centers, making it highly sensitive to parameter settings. This sensitivity renders it vulnerable to outliers and uneven data distributions. In the context of the AND task, where papers often share similar research topics leading to ambiguous boundaries in the embedding space, the initial cluster centers selected by AP Clustering may be inaccurate, resulting in overall poor clustering performance.

B.4. GNN encoders analysis

The GNN encoder plays a vital role in capturing the graph structural features. In this work, we use a GAT-v (Wang et al., 2019) as our graph encoder. We also compared it with other popular graph encoders such as GAT (Velickovic et al., 2017), GCN (Kipf & Welling, 2016), and GIN (Xu et al., 2018). The experimental results are shown in Table B.16.

Across the four datasets, GAT outperforms GCN in Macro-F1 by -0.3 %, 1.85 %, 1.78 %, and 0 %, respectively. These results indicate that GCN and GAT perform comparably on the AMiner-AND and LAGOS-AND datasets, while GAT significantly outperforms GCN on the WhoisWho-v1 and WhoisWho-v2 datasets. This discrepancy is attributed to differences in the density of relationships within the graphs, with GAT generally demonstrating better performance. In contrast, GIN shows Macro-F1 scores that are 3.27 %, 0.32 %, 6.27 %, and 4.34 % lower than GAT across the four datasets, highlighting a substantial performance gap and suggesting that GIN struggles with the graph-rich scenarios typical of the AND task. Meanwhile, GAT-v outperforms GAT in Macro-F1 by 2.33 %, 1.79 %, 2.22 %, and 0.62 % across the four datasets, respectively. These results indicate that GAT's attention mechanism offers a significant advantage in learning embeddings for the AND task, with GAT-v further improving performance by capturing higher-order neighbor features. This demonstrates that higher-order neighbor information is crucial for enhancing disambiguation performance.

Table B.15
Comparison of clustering methods (%).

Model	AMiner-AND			WhoisWho-v1			WhoisWho-v2			LAGOS-AND		
	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1
OPTICS	**85.17**	64.21	69.38	86.57	87.50	85.80	**93.24**	90.82	91.27	88.77	74.64	79.10
DBSCAN	82.46	68.48	**71.43**	86.67	87.62	85.92	93.22	90.83	91.26	89.00	74.78	79.28
AP Clustering	65.60	56.59	55.34	70.37	70.50	67.65	78.34	79.21	77.56	71.18	65.19	64.95
HDBSCAN	81.18	**68.65**	71.24	**87.02**	**89.96**	**87.46**	92.60	**93.45**	**92.57**	**90.52**	**85.71**	**86.12**

Table B.16
Comparison of GNN encoders (%).

Model	AMiner-AND			WhoisWho-v1			WhoisWho-v2			LAGOS-AND		
	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1
GCN	77.52	69.22	69.21	84.27	87.06	83.85	90.70	89.09	88.57	89.72	85.28	85.50
GIN	71.16	**69.53**	65.64	83.31	**91.43**	85.35	83.19	88.43	84.08	87.96	78.90	81.16
GAT	78.49	67.49	68.91	84.08	90.56	85.67	89.56	92.86	90.35	90.22	85.16	85.50
GAT-v	**81.18**	68.65	**71.24**	**87.02**	89.96	**87.46**	**92.60**	**93.45**	**92.57**	**90.52**	**85.71**	**86.12**

Table B.17
Comparison of Semantic encoders (%).

Model	AMiner-AND			WhoisWho-v1			WhoisWho-v2			LAGOS-AND		
	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1
FastText	80.68	67.27	69.39	**87.71**	87.23	86.19	93.74	90.09	90.97	89.10	83.86	84.15
GloVe	74.22	**73.81**	69.54	84.09	**90.26**	85.37	92.71	91.64	91.31	83.02	**88.47**	83.23
ELMo	79.87	68.20	69.86	86.67	86.18	85.07	**93.80**	89.04	90.24	85.73	86.26	85.24
Word2Vec	**81.18**	68.65	**71.24**	87.02	89.96	**87.46**	92.60	**93.45**	**92.57**	**90.52**	85.71	**86.12**

B.5. Semantic encoders analysis

We use Word2Vec as the semantic encoder to encode the initial embeddings of paper nodes in the graph, which serve as inputs to the GNN encoder. Additionally, we conduct a comprehensive comparison of the performance of Word2Vec and other commonly used word embedding encoders in the AND task, including FastText (Joulin et al., 2016), GloVe (Pennington, Socher, & Manning, 2014), and ELMo (Peters et al., 2018). The results are shown in Table B.17.

The experimental results indicate that the initial embeddings generated by different semantic encoders do not have a significant impact on the overall model performance. In terms of the F1 score across the four datasets, FastText performs 1.85 %, 1.27 %, 1.6 %, and 1.95 % worse than Word2Vec, respectively. GloVe performs 1.7 %, 2.09 %, 1.26 %, and 2.89 % worse than Word2Vec, respectively. ELMo performs 1.38 %, 2.39 %, 2.33 %, and 0.88 % worse than Word2Vec, respectively. We analyzed several potential reasons. Although FastText can capture features such as word roots, affixes, and spelling variations, which may perform well in languages with rich morphology, in the AND task, we train the semantic encoder using text from the entire dataset, and during inference, out-of-vocabulary (OOV) words almost never occur. Therefore, the advantages of FastText are not leveraged, and the additional parameters introduced by computing subword information may require a larger dataset to yield better results. GloVe combines global word statistics and local context, which intuitively might lead to better performance. However, this global statistical information may have a negative impact. We use short texts such as titles, keywords, and organization information as input to the semantic encoder, and during training, we also incorporate abstracts, which serve as the primary training corpus. Consequently, the local co-occurrence patterns of these short texts may not be adequately captured by the global statistics, leading to the loss of some details. ELMo is more adept at handling long-text context, but when used with short texts such as titles, keywords, and organization information, where the contextual relationships are weak, it cannot fully leverage its strengths. Additionally, the semantic features are derived from fixed terms rather than dynamic context, and the complex encoding of ELMo may introduce noise. Word2Vec outperforms the others likely because it focuses more on local context. Titles, keywords, and organization information somewhat represent the research direction and affiliation of a researcher, so papers with similar contextual representations are more likely to be similar. The simplicity and local contextual capture of Word2Vec may make it better suited for the AND task. Additionally, in our experiments, we observed that Word2Vec trains significantly faster than the other three encoders, with FastText coming in second, and ELMo requiring a much longer training time.

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